Marine Pollution Bulletin 202 (2024) 116273

Contents lists available at ScienceDirect

Marine Pollution Bulletin

journal homepage: www.elsevier.com/locate/marpolbul

Review

# A systematic review of robotic efficacy in coral reef monitoring techniques

Jennifer A. Cardenas [a], Zahra Samadikhoshkho [a], Ateeq Ur Rehman [a], Alexander U. Valle-Pérez [b,c,d], Elena Herrera-Ponce de León [b,c,d], Charlotte A.E. Hauser [b,c,d], Eric M. Feron [b], Rafiq Ahmad [a,*]

[a] Aquaponics 4.0 Learning Factory (AllFactory), University of Alberta, Edmonton, Canada
[b] Laboratory for Nanomedicine, Division of Biological and Environmental Science and Engineering, King Abdullah University of Science and Technology, Thuwal 23955-6900, Saudi Arabia
[c] Computational Bioscience Research Center (CBRC), King Abdullah University of Science and Technology, Thuwal 23955-6900, Saudi Arabia
[d] Red Sea Research Center, King Abdullah University of Science and Technology, Thuwal, Jeddah 23955, Saudi Arabia

ARTICLE INFO

*Keywords:*
Monitoring
Artificial intelligence (AI)
Bio-sensors
Coral reefs
Coral bleaching
Red Sea
Robotics
Autonomous underwater vehicles (AUVs)
Marine conservation
Ocean monitoring
Vertical farming

ABSTRACT

Coral reefs are home to a variety of species, and their preservation is a popular study area; however, monitoring them is a significant challenge, for which the use of robots offers a promising answer. The purpose of this study is to analyze the current techniques and tools employed in coral reef monitoring, with a focus on the role of robotics and its potential in transforming this sector. Using a systematic review methodology examining peer-reviewed literature across engineering and earth sciences from the Scopus database focusing on "robotics" and "coral reef" keywords, the article is divided into three sections: coral reef monitoring, robots in coral reef monitoring, and case studies. The initial findings indicated a variety of monitoring strategies, each with its own advantages and disadvantages. Case studies have also highlighted the global application of robotics in monitoring, emphasizing the challenges and opportunities unique to each context. Robotic interventions driven by artificial intelligence and machine learning have led to a new era in coral reef monitoring. Such developments not only improve monitoring but also support the conservation and restoration of these vulnerable ecosystems. Further research is required, particularly on robotic systems for monitoring coral nurseries and maximizing coral health in both indoor and open-sea settings.

## 1. Introduction

### 1.1. Coral reef monitoring

Coral reefs are critical for marine biodiversity and worldwide income; they are the habitat of a diverse range of marine life, and provide significant commercial and cultural opportunities, such as tourism (Snow et al., 2020; Rogers et al., 1994; Mulhall, 2004). Despite covering a tiny fraction of the ocean surface and mostly being in tropical regions, coral reefs are the primary habitat for almost a quarter of all marine species (Mulhall, 2004). To address the significant threats to coral reefs, from climate change to overfishing and pollution, this study is focused on the research question: how can the development of new methodologies in coral reef monitoring, particularly through the use of robotics, contribute to more effective, efficient, and robust conservation efforts? This section explores the status and the challenges of coral reefs; providing a discussion of advanced technologies that are being implemented for their conservation and monitoring, with a focus on robotics.

(Snow et al., 2020; Rogers et al., 1994; Mulhall, 2004) Unfortunately, coral reefs face threats from natural disturbances and human activities such as urban development, overfishing, ocean acidification, industrial pollution, invasive species, and climate change (Alonso et al., 2019), causing an alarming decline in their growth, health, and overall ecosystems (González-Rivero et al., 2016). Some additional causal factors affecting the coral reef ecosystem include the crown of thorn starfish outbreaks (Fabricius et al., 2010), bleaching and mortality of coral (Baker et al., 2008; Hoegh-Guldberg, 1999), tropical storm destruction (De'Ath et al., 2012), and the impacts of rising sea surface temperature, sedimentation, eutrophication, and nutrient runoff (Brodie et al., 2005). To understand the rapid decline of coral reef ecosystems, the causes must be identified (Hughes et al., 2010; Knowlton and Jackson, 2008).

In the last 40 years, coral cover has decreased by more than 80 % in

* Corresponding author.
*E-mail address:* rafiq.ahmad@ualberta.ca (R. Ahmad).

https://doi.org/10.1016/j.marpolbul.2024.116273
Received 5 October 2023; Received in revised form 15 March 2024; Accepted 16 March 2024
Available online 3 April 2024
0025-326X/© 2024 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

the Caribbean (Gardner et al., 2003), and between 1997 and 2003, anywhere from one to 2 % annually in the Indo-Pacific region (Gardner et al., 2003). It is anticipated that these loss trends will increase (Van Hooidonk et al., 2013). However, global temperatures are expected to rise by 2–4.5 °C by 2100, implying that by 2050, massive annual bleaching of coral incidents is expected to occur (Mulhall, 2004). The richness and number of coral reef fish and animals are affected by habitat loss.

Given the ecological and economic roles of coral reefs in protecting coastlines, attracting tourists, and sustaining fisheries (Gao et al., 2021) monitoring rapid changes and their health using various biological, chemical, and physical indicators is essential. Monitoring the health and condition of coral reefs is crucial for several reasons.

Understanding the highly dynamic environment of coral reefs, including changes in water temperature, dissolved oxygen levels, and pH, is crucial for protecting these ecosystems (Gao et al., 2021). Many surveys are required to learn more about the "coral reef crisis" (Hughes et al., 2003), informing ecosystem-based management approaches for safeguarding these ecosystems. Additionally, monitoring coral reef ecosystems over the course of several years helps develop effective management practices and strategies. This includes international efforts like the Coral Reef Conservation Act and the initiatives under the International Coral Reef Initiative (McCook et al., 2010; Mills et al., 2010). However, surveying coral reefs requires specialized technical knowledge and substantial financial commitment. In addition, because coral reef access is remote and monitoring requires scuba diving, long-term records are often scattered or limited to spatial datasets (Hughes et al., 2010).

### 1.2. Education and research

Despite comprising nearly 70 % of the Earth's surface, the oceanic environment has been completely ignored (Visbeck, 2018). This section discusses the challenges unique to marine research, such as adverse environmental factors, lack of adequate infrastructure, and higher operational costs compared to terrestrial and aerial studies (Hammad et al., 2023). Subsequently, different coral reef monitoring methods are explored, both traditional and technological, and their implications for research and education in marine ecology.

As shown in Fig. 1, different monitoring methods are used to monitor coral reefs, involving both biological aspects and physical environmental conditions, including: in-situ observations such as diver-based monitoring, remote sensing, and autonomous and semi-autonomous monitoring. Below, we explain some of these methods in detail:

- Diver-based Monitoring

Traditionally, reef surveys have been carried out by scuba divers as the most robust monitoring approach, with expertise in marine ecology that can identify and count taxa while exploring underwater reefs (Gray et al., 2016). The ability of the divers to navigate and adapt to the dynamic underwater environments enables the collection of meticulous observations that automated systems might miss. Moreover, in environments where the fragility of coral ecosystems requires gentle handling, the abilities of human divers are irreplaceable. Although in situ surveys yield precise observation data, they can only cover a small portion of the reef and take considerable time to complete. The original material is often unavailable for further analysis and relies heavily on the proficiency of surveyors. Scientists regularly dive into reef locations to visually evaluate their state of health. Since scientists may need to travel considerable distances to reach different locations with living coral, this approach is time-consuming and tedious (Chennu et al., 2017). Planning each dive in advance based on aerial photography by locating possible reef areas is an additional strategy for dealing with this problem. However, its effectiveness relies on satellite photos with sufficient visual clarity and resolution (Beijbom et al., 2015; Shkurti et al., 2012).

However, several risks could arise from human intervention in the marine environment, including drowning, decompression illness, collisions with ships, and not permitted site access. Additionally, there is a strong likelihood that the effort will need to be repeated, and owing to the biological constraints of the human organism, less time is available for exploration (Aumeer et al., 2021). The complexity of these studies can occasionally threaten human lives. Observation of the Veracruz Reef System in Mexico is a situation where the depth is greater than 30 m, and a navigation channel threatens the safety of investigating divers.

When affordable and high-quality underwater cameras became widely available in the early 1960s (Jaffe et al., 2001), they mainly improved the methods available for monitoring coral reefs, complementing in situ surveys. Using images to conduct surveys has many benefits, including the quick collection and efficient observation of data on the state of coral reefs and the preservation of records that can be mined for information on population density and the distribution of specific species (Pante and Dustan, 2012; Gleason et al., 2010). However, compared to in situ inspection, estimating species in benthic pictures is more difficult because of the low resolution of images, variable illumination conditions, water clarity, and difficulty engaging directly with the benthos. There is a lack of evidence detailing the variation brought about by coral reef survey photo annotations, despite some efforts to analyze the precision and inter-observer variation of underwater video transect surveys (Carleton and Done, 1995; Ninio et al., 2003).

Three major advancements have been made in the use of image-based technologies in marine ecology. First, developments in digital photography have made it possible to collect massive amounts of data with ever-improving resolutions. Second, computers and software have

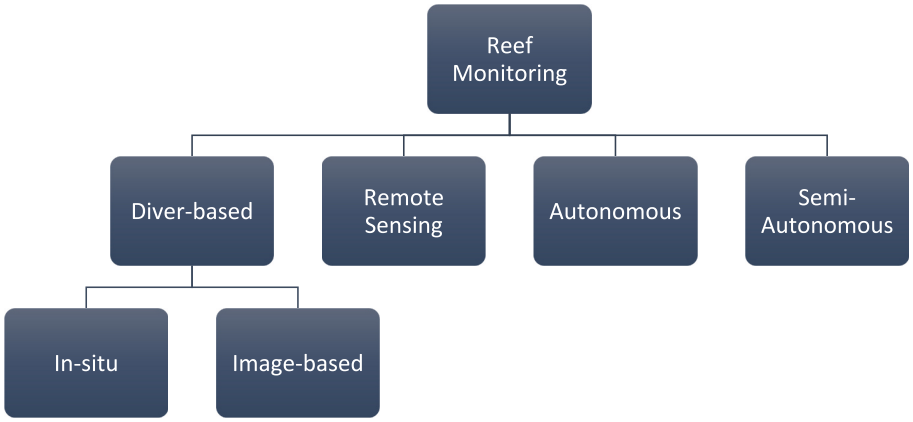


**Fig. 1.** Reef monitoring methods, adapted from (Obura et al., 2019)

vastly improved their capacity to process, store, and analyze visual data. Third, technological developments in automation and robotics have enabled the creation of autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), and imaging sleds capable of capturing hundreds of survey images in a single arrangement (Singh, 2004; Nicholson and Healey, 2008; Eleftheriou, 2013; González-Rivero et al., 2014). Quantitative ecological information is often generated through the random point annotation of underwater photographs to estimate the percentage cover of the surfaces of interest. Substratum types are determined for a random selection of sites within each image to generate this data, which is often performed manually by an expert in that environment. Careful consideration of the study's primary purpose, for a statistically valid sampling of benthic habitats with image- and point-based tools, is important to choose the experimental and statistical methods needed to answer the questions at hand, as well as the statistical power (which is a function of sample size, variance, and the size of the difference you want to find).

- Remote Sensing Monitoring

Drawing broad conclusions from marine ecological observations might be difficult due to regional heterogeneity. There is a need for high-resolution images and standardized and georeferenced field information, even though improvements in integrative modeling, meta-analysis, and remote sensing allow for generalization from field monitoring. Meta-analysis has shown promise as a tool for synthesizing underwater assessment efforts and making broad regional generalizations (Gardner et al., 2003; Schutte et al., 2010; Hughes et al., 2002; Parmesan and Yohe, 2003; Côté et al., 2005), but it can be difficult to make such claims with confidence due to factors such as spatial scale, observer bias, metrics, and methodology. Researchers may be able to scale up field observations and gain a better understanding of the change-driving mechanisms in coral reefs by employing multidisciplinary and collaborative strategies based on comprehensive field data collection, modeling techniques, and optical remote-sensing information sets (satellite- and aircraft-derived images). (De'Ath et al., 2012; Scopélitis et al., 2007; Mumby et al., 2007; Roelfsema, 2010). Advantages of remote sensing include a quickly vast areas coverage and data on a large scale, making it invaluable for tracking changes over time across extensive coral reef systems. This method is particularly effective in inaccessible regions or where human presence is limited, ensuring minimal disturbance to the natural habitat.

However, despite these benefits, remote sensing has limitations in two main areas: resolution and specificity, where the detail it can provide is often insufficient for identifying specific coral species or assessing the health of individual corals. Regardless of the method chosen, comprehensive and standardized field data are essential for keeping track of reefs in transition and for making recommendations for management that are prompt and accurate (Hughes et al., 2010). Optical remote sensing can scan entire coral reef systems (e.g., 10,000 $km^2$) from the air with 100 % coverage (Mumby et al., 2004). Each pixel was evaluated in relation to its habitat composition, and the size of the pixel determined the level of benthic information recorded. Compared to measurements taken in the field, optical remote-sensing products fall short of detail and accuracy (Hedley et al., 2012). In contrast, field-based observations often only cover small sections of the study site (Edmunds et al., 2018) and are crucial for generated map validation and remote sensing image calibration (Roelfsema, 2010; Mumby et al., 2004; Hedley et al., 2012; Wooldridge and Done, 2004).

- Autonomous and Semi-autonomous Monitoring

Reef monitoring and data collection from the environment are crucial and expensive tasks. Numerous agencies have used remote observation stations and divers to undertake this task at a high cost (Lawrance et al., 2019). These organizations created methods to raise data collection quality and rates to improve monitoring by obtaining a broader perspective instead of inferred views from fewer data streams. However, these technologies are typically launched by operators or are remotely controlled. Because most existing monitoring programs lack significant autonomy, it is evident that more can be done to increase data collection rates while decreasing monitoring expenses (Dunbabin et al., 2005).

To address this issue, the development and implementation of autonomous robotic devices has been proposed. These vehicles were classified into two categories based on their autonomy levels. Remotely operated vehicles (ROVs) require constant human control, whereas autonomous underwater vehicles (AUVs) provide some level of automatic operation without the need for a human operator (IEEE Staff, 2016). Autonomous underwater vehicles (AUVs) represent a significant improvement in this respect, facilitating the automation of various tasks previously carried out by manned vehicles in underwater settings (González-García et al., 2020). AUVs have been intensively explored, and their capabilities and uses continue to evolve. However, the expense of AUVs continues to limit their availability to research institutes and large business users (Lawrance et al., 2019).

Low-cost remotely operated vehicles (ROVs) costing less than $50,000, on the other hand, have become more available. ROVs have been noted for their toughness and task adaptability, making them ideal for a variety of time-critical operations (Lawrance et al., 2019). However, the human operation of ROVs presents significant challenges, as it demands a high degree of operator training and skill, while the constant attention required can be draining for operators (IEEE Staff, 2016).

### *1.3. The importance of interdisciplinary collaboration in addressing coral reefs' global challenges*

Due to their multifaceted nature, environmental challenges are typically difficult to manage within a single discipline. They necessitate insights from multiple disciplines and the collaboration of experts from various fields, including biology, chemistry, physics, engineering, economics, politics, and the social sciences. This section recognizes and states the need for a multifaceted approach in order to address the challenges faced by coral reefs. For instance, the integration of materials engineering, electric engineering, bioengineering, and marine sciences has resulted in the development of eco-friendly and sustainable biomaterials for 3D coral printing and molding (3D CoraPrint) as well as the integration of machine learning and 3D bioprinting for the development of novel emerging approaches for coral restoration (Hammad et al., 2023; Wangpraseurt et al., 2022; Avila-Ramírez et al., 2021; Albalawi et al., 2021). Furthermore, the integration of AI, machine learning, and neural network programming can enhance coral restoration strategies and technologies. This collaboration can bridge the gaps between different disciplines to address the associated environmental issues, including conservation planning (Knight et al., 2008), resource management (Shackleton et al., 2009), restoration, and invasion management by providing rounded and broad information, comprehensive theories, methods, and the philosophical perspective employed by any discipline (Szostak, 2013). Integrating concepts or methods from several social and natural sciences, and possibly including stakeholders outside academia, is important in finding solutions to these problems (Macleod and Nagatsu, 2018). The use of robotics in coral reef monitoring leads to improved accuracy, efficiency, and cost-effectiveness in data collection compared to traditional monitoring methods. The aim of this study is to evaluate the effectiveness of robotics in coral reef monitoring across different areas including data accuracy, operational efficiency, and economic feasibility. To identify challenges and opportunities in the implementation of robotic technologies for coral reef conservation.

## 2. Research methodology

This study conducts a systematic review methodology. To assess the

current state of coral reef monitoring research, with a particular focus on the role of robotics, we systematically reviewed the literature. For the data acquisition, Scopus database was selected as the primary database for its extensive coverage of literature in the fields of engineering and earth sciences, which are crucial for this cross-disciplinary study. The data was retrieved through the option of title/abstract/keywords, using the keywords "robotics" and "coral reef"; this being limited to the subject areas of Engineering and Earth Sciences with the purpose of a selection of only relevant literature. Only peer-reviewed English journals and conference proceedings were considered for the review procedure. Book reviews and editorials were eliminated for review purposes. After the consideration of the established criteria for the formula, duplicates removal, a total of 77 research articles were obtained for the systematic analysis for this study, which initially recovered 88 documents. After retrieval, the study was separated into three major sections: coral reef monitoring, robots in coral reef monitoring, and case studies. This allowed us to identify and analyze the many techniques and tools utilized for coral reef monitoring, and to evaluate the efficacy of these tactics in terms of accuracy, cost-effectiveness, and convenience of use. Fig. 2 illustrates the data collection and analysis procedures used in our research.

A systematic review is a research study in which all available evidence on a topic or question is identified, selected, and synthesized. This method provides a thorough and objective examination of the available literature, which is beneficial for identifying gaps in knowledge, inconsistencies in findings, and topics for future research (Munn et al., 2018). Our review focused on three main research areas: monitoring coral reefs, robotics in coral reef monitoring, and a case study.

In the first area, we sought to identify and evaluate the different methods used to monitor coral reefs, including the importance of monitoring coral reefs and the available monitoring tools and techniques. We also examined the limitations and advantages of this study. We focused on the use of robotics for coral reef monitoring in the second area. We identified studies that have explored the use of underwater robots and other autonomous vehicles to collect data on coral reefs. We analyzed the effectiveness of these technologies in terms of accuracy, cost-effectiveness, and ease of use as well as their potential to revolutionize coral reef monitoring in the future. We also identified the advantages and limitations of using robots for coral reef monitoring, as well as artificial intelligence and machine learning for this purpose.

Finally, in the third area, we analyzed a range of case studies that documented the applications of robotics in coral reef monitoring worldwide. We examined monitoring techniques in specific applications and the challenges and opportunities faced. We also assessed the effectiveness of these strategies in terms of their ability to monitor and protect coral reefs accurately. Furthermore, we discuss the implications of these studies for future research and conservation.

## 3. Monitoring coral reefs

### 3.1. Importance of monitoring coral reefs

Monitoring coral reefs is crucial for preventing damage from various threats to secure their long-term benefits (Hedley et al., 2016a). However, coral reefs are at risk of decline. To protect coral reefs and ensure their continued benefits, it is essential to establish standard operating procedures (SOPs), policies, and conservation rules. These measures include 1) establishing protected areas, 2) implementing conservation policies, 3) developing SOPs, and 4) promoting public awareness and education (Hughes et al., 2010). Thus, monitoring coral reefs across the globe is a crucial component of coral reef conservation. Regular monitoring can help us understand how coral reefs change over time; identify threats of mass bleaching, disease outbreaks, and vulnerabilities; and evaluate the effectiveness of conservation measures (Gastoldi and Cinti, 2023; Obura et al., 2019).

### 3.2. Available monitoring tools and techniques

Various tools and techniques can be used to monitor the health and status of coral reefs, including the use of scuba diving or snorkeling equipment to visually assess the health, diversity, and abundance of corals and other marine habitats (Teague et al., 2022). This technique can provide a wealth of information on the condition of the reef and its inhabitants, including coral cover, coral growth rates, and disease or coral bleaching (Safuan et al., 2022). Coral health assessments involve a detailed examination of coral colonies to identify signs of disease, bleaching, or the effects of other stressors. This technique can help researchers to understand the underlying causes of coral decline and develop conservation and restoration strategies (Giguere et al., 2009). Satellite remote sensing is another method used by researchers to monitor changes in sea surface temperature, water clarity, and other environmental factors that can impact coral health. This technique can help identify areas of the reef that are at risk of bleaching or under the threat of other stressors. This method has been used to develop early warning systems for coral reef conservationists (Hoogenboom et al., 2012; Hedley et al., 2016a; Safuan et al., 2022).

Acoustic monitoring involves the use of underwater microphones to record the sounds of reef organisms such as fish and invertebrates. This

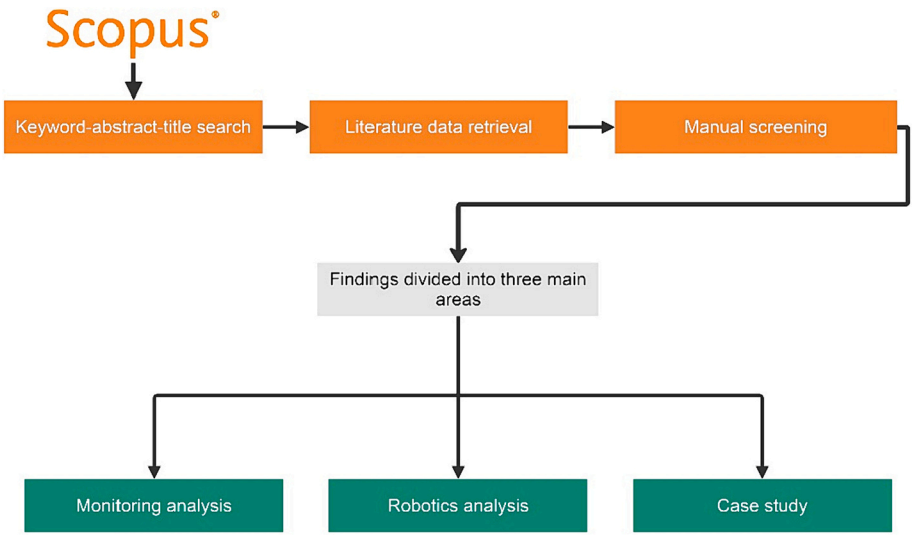


**Fig. 2.** Systematic review.

technique can provide insights into the abundance and diversity of different reef species and the overall health of the marine ecosystem (Zamani et al., 2022). Environmental DNA (eDNA) analysis involves DNA sequencing to detect the presence of different species in water samples taken from the reef. This technique can be used to identify rare or elusive species that are difficult to detect through visual surveys (Lau et al., 2019). Citizen science initiatives such as Reef Check involve training volunteers to conduct visual surveys and collect data on coral reefs. This approach can help expand the scope of monitoring efforts and engage local communities in conservation and restoration efforts (Gouraguine et al., 2019; Hughes et al., 2010). These tools and techniques can be used individually or in combination to monitor the health and status of coral reefs and to inform conservation and restoration efforts (Safuan et al., 2022).

### 3.3. Limitations of current monitoring methods

Coral reefs face unique challenges (Kalantari et al., 2020) and require specialized monitoring methods (Girdhar et al., 2019). Coral reefs are diverse and complex ecosystems (Murad et al., 2016), and traditional monitoring methods can only sample small areas or specific types of coral. This can result in limited information regarding the overall health and resilience of coral reefs (Hughes et al., 2010; Madin et al., 2019). Accessing coral reefs in remote areas makes regular monitoring challenging (Osborne et al., 2011). This can limit the monitoring frequency, making it difficult to detect changes or respond quickly to threats. Another limitation of monitoring coral reefs is the expense of monitoring equipment, particularly for long-term monitoring programs that require regular surveys and data analysis. This can limit the availability of monitoring resources, particularly in developing countries where coral reefs are the most vulnerable (González-Rivero et al., 2020; Girdhar et al., 2019). Corals are sensitive to disturbances (Edmunds et al., 2018), and traditional monitoring methods can damage the reef or stress corals. This can affect the health of corals and limit the accuracy of the data collected.

Although considerable data are available on coral reefs, there are still many gaps in our understanding of the complex ecological relationships in these ecosystems (Morais et al., 2018; Maslin et al., 2021). This can limit the effectiveness of monitoring methods, particularly if important variables are not included in the monitoring programs. Coral reefs change slowly over time (Berkelmans and Van Oppen, 2006), and it can be difficult to detect subtle changes in reef health. This can require long-term monitoring programs, which can be challenging to maintain over time. Therefore, focused monitoring methods and tools are necessary to overcome some of the limitations of traditional monitoring tools. Addressing these limitations requires specialized monitoring methods and tools that account for the unique characteristics of coral reefs. This may include new technologies, such as robotics (Wang et al., 2022; Hogue and Jenkin, 2006) and underwater sensors (Sattar et al., 2009; Zhang and Negahdaripour, 2003). In Table 1, a comparative summary of each previously discussed monitoring method is illustrated. By using specialized monitoring methods, scientists, researchers, and conservationists can gain a more comprehensive understanding of the health of coral reefs and identify areas that require targeted conservation efforts.

**Table 1**
Summary of comparative analysis of coral reef monitoring methods.

| Monitoring method | Advantages | Limitations | Recommendations |
|---|---|---|---|
| Visual surveys (Pizarro et al., 2017; Wetz et al., 2020) | Direct assessment of the coral's reef health; providing detailed information on coral cover and growth rates. | Limited to the depth range of the diver, time-consuming and represents a potential reef damage. | Mainly for easily accessible coral reefs. It can be combined with other methods for a more extensive analysis. |
| Coral health assessment (Safuan et al., 2022; Manderson et al., 2017a; Manderson et al., 2016) | Help to understand the causes of coral declines. | Requires expert knowledge and is limited to accessible areas. | Focused on specific coral health issues. |
| Satellite remote sensing (Hoogenboom et al., 2012; Safuan et al., 2022) | Identification of areas at risk of bleaching; being useful for early warning systems. | Lower resolution in comparison to methods in-situ. | Ideal for a broad monitoring of coral reef's ecosystem health. |
| Acoustic monitoring (Zamani et al., 2022; Katzschmann et al., 2018a; Dimoff et al., 2021) | Non-intrusive. | Limited to acoustic properties without visual data. | Suitable for ongoing monitoring of the health of coral reefs ecosystems. |
| Environmental DNA (eDNA) Analysis (Zamani et al., 2022) | Detect rare species, non-intrusive and provides detailed biodiversity data. | Requires specialized equipment and expertise. Additionally, it does not provide real-time data. | Idea for biodiversity studies. |
| Citizen science Initiatives (Lau et al., 2019; Gouraguine et al., 2019) | Engage local communities, cost-effective. | The accuracy of the data is variable, depending on the training of the volunteers. | Good for increasing public awareness and involvement in the coral reef conservation. |
| Robot monitoring (Snow et al., 2020; Shkurti et al., 2012; Dunbabin et al., 2006; Avila et al., 2021a; Dunbabin et al., 2004; Girdhar and Dudek, 2016; Biswas et al., 2021) | Accurate data collection, real-time data, non-intrusive. | High cost of initial investment and requires technical expertise. | Recommended for long-term in-depth monitoring and areas that are not accessible to divers. |

### 3.4. Advantages of using robots for coral reef monitoring

Robots can be used for various tasks related to coral reef monitoring and research (Katzschmann et al., 2018b). There are numerous advantages of using robots for coral reef monitoring. Robots can collect data from coral reefs more accurately and precisely than human divers (Obura et al., 2019). Robots can collect and transmit data in real-time. Previously, heterogeneous robot systems were used to collect underwater data to monitor coral reefs, where the underwater robots were connected first with surface robots and then with aerial robots to transmit the data, allowing researchers to analyze the data remotely and make decisions more quickly than if they had to wait for divers to return to the surface (Duong et al., 2020). Robots can operate non-intrusively by taking a series of images with subsequent analysis, meaning that robots can collect data without causing damage to the coral or disturbing marine life inhabiting the reef (Cambria and White, 2014). This is particularly important for delicate coral ecosystems that are already threatened by climate change and other human activities (Berlian et al., 2017).

However, the use of robots for coral reef monitoring also presents disadvantages including the initial cost for developing, deploying, and maintaining robotic systems which can prevent their use in research institutions and conservation projects (Behmanesh et al., 2017). Additionally, robots may have limitations in accessing very complex underwater structures compared to human divers, potentially missing critical

data in these environments (Lam et al., 2007). Moreover, the technology also requires specialized skills for operation and maintenance, which can add to the overall cost and complexity of research projects (Avila et al., 2021b). In certain scenarios, the presence of robots, especially those not designed for minimal disturbance, can still affect marine life behavior (Venkatesh et al., 2013). Using robots for coral reef monitoring can be more cost-effective in the long term than sending human divers to collect data. While the initial investment in robot technology may be high, maintaining, and operating robots over time can be significantly lower than employing a team of divers (UK-RAS Network Robotics and Autonomous Systems, 2018). Taken together, using robots for coral reef monitoring has the potential to provide more accurate and detailed data while minimizing the impact on coral ecosystems and reducing the costs and risks associated with human divers.

## 4. Robotics in coral reef monitoring

The application of robotics has expanded to numerous engineering and environmental disciplines. Monitoring the environment and identifying diseases are just two examples of the many uses of underwater and agricultural robots (Behmanesh et al., 2017). In coral reefs, changes happen over a wide range of sizes, from millimeters to kilometers, and the way dominant species are zoned and grow makes clear patterns. Coral reef studies have found it difficult to study both complicated biological systems and structures that change as coral grows and dies. This obstacle shows the importance of using a cross-scale, highly automatic method.

The physical constraints the human body imposes significantly restrict what can be discovered. This heritage of exploration can now be carried out without endangering human lives thanks to robotics (Girdhar and Dudek, 2016). Robots can be used in different applications to preserve coral reef ecosystems, such as observing coral health and growth (Lam et al., 2007), analyzing the behavior of reef organisms, detecting coral bleaching, identifying invasive species, monitoring their spread, and collecting data on water quality and other chemicals.

To monitor coral reefs at varying depths, (Avila et al., 2021b) built a robot that can operate underwater using an artificial vision system. This robot can switch between several operational modes to complete its missions, including teleoperation for surface motion and underwater video access.

### 4.1. Overview of the use of robotics in coral reef monitoring

Cooperative multi-agent systems and intelligent robots can be valuable for expediting research in remote areas. Robots can also perform difficult or impossible tasks because they are in hazardous settings or situations. Underwater robots seek the technology and information underlying the analysis of underwater video footage, detection of underwater features, pose estimation, vehicle control, mechanical design, and human-robot interaction. Assessing marine ecosystems and the diversity of life in coral reefs are two critical applications of underwater robots (Venkatesh et al., 2013).

The accuracy of visual observations for identifying multiple coral reefs from numerous species is highly dependent on dive time span, depth, and water clarity (Villon et al., 2018). Therefore, Underwater Vehicles (UAVs) can help individuals to discover the sea environment in new ways. These carefully made robots can remain submerged for significantly longer than divers, giving scientists more time to explore. Divers that go underwater must be trained and have expensive equipment to help them breathe. On the other hand, UAVs are easy to use and only need to be purchased once.

Robotic advancements have made it easier to collect data under difficult conditions, such as aerial (Koh and Wich, 2012) and underwater (Bryant et al., 2017; González-Rivero et al., 2014) surveys. For instance, vision sensors are a commonly used technology that generate enormous amounts of data with little technical knowledge. Researchers can move the bottleneck in extensive ecological studies and surveillance from image acquisition to image processing by quickly documenting vast areas using high-resolution photos. The manual process of extracting valuable data from these datasets is laborious and necessitates innovative methods and automation. Before the information is ready to be used to guide conservation decisions, it frequently takes a substantial amount of time to analyze the digital information in each image to provide environmentally relevant metrics, such as benthic structure. Due to this delayed effect, there is a significant bottleneck in the flow of information between monitoring programs and conservation researchers (González-Rivero et al., 2020).

Time and resource constraints associated with data acquisition and collection may prevent the advancement of precise models of marine systems and act as bottlenecks (González-Rivero et al., 2020). Long-term ocean missions to gather water samples or survey benthic oceans can be expensive and time consuming (Wölfl et al., 2019). Even on coastlines, which are often more accessible than the open ocean, restrictions on the supporting infrastructure of the deployment area can impede continuous data collection and monitoring (Janzen et al., 2019). It has been demonstrated that robotics system innovations facilitate data acquisition and alleviate some of these concerns (Wölfl et al., 2019; Ruiz et al., 2009).

Space and ocean research requires the employment of autonomous robots due to the severe communication bottlenecks that prevent vehicles from direct remote control. However, the intrinsic duration of such exploration missions necessitates autonomy beyond the basic navigational control. Real-time and sophisticated comprehension of the environment is required to increase the richness of the acquired data for a mission. This can then be used to plan the path of the robot adaptively.

Underwater robots are equipped with GPS systems and different sensors such as cameras, inertial measurement unit (IMU), pressure, among others; that allow the data collection from their surroundings (Poongundran et al., 2022). In this sense, one of the largest challenges that is faced when designing underwater robotics is reliable wireless communication, complicating the data transfer and storage. During underwater explorations, the environment has a negative impact on transmission methods, including acoustic, electromagnetic, magnetic induction, optical communications, and limited radio range, which results in low data range, interference, and link disruptions (Kirchner et al., 2020; Joordens and Champion, 2015). This represents a challenge for the further development and scalability of AI and IoT systems in underwater robotics (Ditria et al., 2022).

### 4.2. Types of robots used in coral reef monitoring, including stationary and mobile biosensors

Coral reef monitoring employs various types of robots for data collection and research assistance. Some of the basic categories of such robots are shown in Fig. 3.

#### 4.2.1. Remotely operated vehicles (ROVs)

Diver-based data collection in field surveys has limitations regarding dive duration, water temperature, weather conditions, and area covered. Additionally, inexperienced divers can damage the reef ecosystems. To replace these methods, remotely operated vehicles (ROVs) have been replaced with human divers, particularly in deep-sea conditions, where divers cannot operate safely. Two important applications of the ROV technology are monitoring the environment in shallow water and reef surveys. Owing to the expensive nature of previous ROV technologies, only a small portion of the underwater regions of the world has been studied (Lam et al., 2006). In hazardous or deep environments, inspection ROVs that operate with surface operators can replace divers. ROVs improve the efficacy of submerged tasks by continuously operating and transmitting video feeds and scientific data to surface operators (Capocci et al., 2017).

Inspection ROVs can be classified as medium, portable, or miniature.

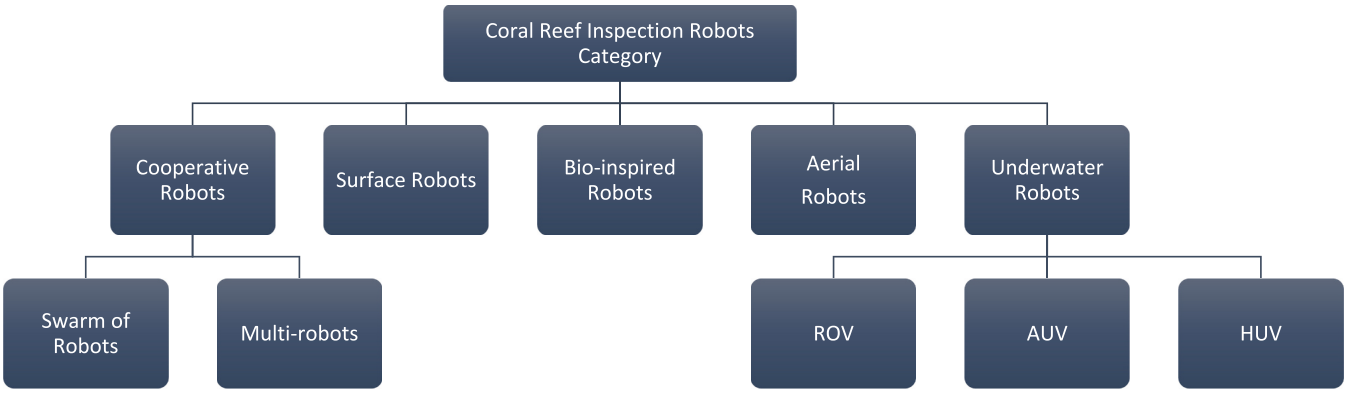


**Fig. 3.** Coral reef monitoring robots category, adapted from (Obura et al., 2019).

Medium-sized inspection ROVs can be installed and recovered manually by personnel, whereas larger vehicles may require an expensive Launch and Recovery System (LARS). Most medium-sized ROVs in the inspection category have open-frame designs, enabling additional sensor installation and tiny device platform additions. In addition to inspections, some vehicles can clean, sample, and recover coral reef ecosystems. Medium-sized inspection ROVs can be used to map and survey underwater with accurate navigation and high-resolution imaging. There is a possibility for imaging sonars to be deployed independently of navigation systems and employed as real-time "acoustic eyes" for navigation in unclear seas. For accurate sonar surveys, an open-frame ROV is more stable. With high thrust, ROVs can overcome their massive volume and drag, allowing for good control in adverse situations. Miniature inspection ROVs can be installed and recovered by operators. These ROVs reduce operational expenses and system complexity, making them efficient. Handheld ROVs can be cube shaped or streamlined. Their design makes them less stable than open-frame medium inspection ROVs. Handheld thrusters have lower power and volume than medium-sized inspection ROV thrusters. Despite their modest mass, their thrust-to-weight ratio might be large. This class of ROV is mostly used for inspection; however, some have small manipulators for collecting low-weight substances. Several producers use sonar equipment with these vehicles, but they are inappropriate for precise mapping owing to navigational inaccuracies. Therefore, adding additional features may significantly decrease the thrust-to-weight ratios and impair handling and control (Capocci et al., 2017).

Numerous researchers have utilized expensive ROVs and/or acoustic and sonar positioning devices, which require substantial infrastructure to operate effectively. Although this is promising research, their current methods require costly hardware and offline processing to aid localization, which is prohibitive for real-time, large-scale surveying. The use of vision in coral reef environments is regarded as an effective method due to the terrain's wealth of features. Vision equipment is relatively less expensive than other underwater sensors, but requires significantly more data processing to obtain the required information (Dunbabin et al., 2005).

#### 4.2.2. Autonomous underwater vehicles (AUVs)

Fully unsupervised autonomous underwater vehicles (AUVs) are developing and making their way into real-world usage. AUVs are currently used for only a few tasks, but further research and development could make them more useful. Globally, numerous AUVs with diverse functions and missions are operating. AUVs are emerging as the next viable alternative to human explorers for remote monitoring and survey tasks due to recent advancements in computing and energy storage hardware (Dunbabin et al., 2006).

Even in deep waters (Grasmueck et al., 2006), AUVs and surface vehicles have been used for a variety of monitoring tasks, with a focus on coral reef inspections (Beall et al., 2010; Giguere et al., 2009; Singh, 2004). AUVs use sonar and other sensors to map the seafloor in certain situations, providing a more comprehensive view of the underwater environment than conventional methods. However, these methods have limitations such as a rapid rate of data acquisition and a lack of accurate vertical modeling. In addition, the swath widths of high-resolution sensors near the ocean floor are typically narrow. A strategy for addressing this issue entails deploying a group of robots operating in a coordinated formation to increase the sensors' effective swath width and spatial coverage. This necessitates novel techniques for coordinated motion control, distributed decision-making, and intervehicle communication, which are emergent research areas in marine technology (Kalwa et al., 2016). (Sattar et al., 2009) analyzed numerous challenges encountered by autonomous underwater robotics, including interactive communication with underwater personnel, image processing, and hovering vehicle orientation and motion.

With the available technology, an autonomous underwater system is required to observe the in-field environment reliably and efficiently for a much lower cost than the available vehicles (Dunbabin et al., 2005). Most current AUVs employed to investigate coral reefs are costly. (Murad et al., 2016) focused on a cost-effective AUV project that inspected coral reef ecology, which is crucial for preserving natural resources in tropical waters.

As exemplified by the efforts of HOI (Prestero, 2001) and Virginia Polytechnic, the advancement of smaller and less expensive AUVs has received some attention in recent years. However, these vehicles are torpedo-shaped and have restricted perception without vision and maneuverability. This excludes them from coral reef environments. Other larger available AUVs have been found to be prohibitively costly, and their endurance and tethers are deemed restricted. Solar-powered AUV and underwater glider technology are two examples of vehicle designs that explore alternative propulsion and power methods to extend the range and endurance. Although these vehicles have proven successful in ocean trials, their operation in coral reef ecosystems has not been investigated.

#### 4.2.3. Hybrid underwater vehicle (HUV)

Existing practices require human operators to be involved in the operational process of highly demanding situations. This can be accomplished through conventional ROV operations or, more frequently, through hybrid solutions in which the vehicle maintains a direct command-and-control connection (such as a tether) but possesses sufficient autonomy to carry out essential missions in the event of disconnection or to operate on an autopilot with minimal supervision. However, these solutions only address a portion of the operating marine vehicles in complex environments (Kalwa et al., 2016). For example, an HUV based on ROV technology with AUV autonomy was proposed in (Aumeer et al., 2021). The HUV collects and transmits vital coral reef data to a ground station via an Internet of Things (IoT) connection in the suggested system. The collected information is stored in the cloud and

then made accessible via a website and mobile application, allowing immediate online viewing. In addition, artificial intelligence algorithms incorporated into the system enable the automated monitoring of coral bleaching via image processing, as shown in Fig. 4.

The robot in (Snow et al., 2020) uses a partially submerged cable while collecting data autonomously. By restricting the movement of the autonomous system to a single degree of freedom, the system becomes highly capable of navigating coastal environments with minimal effort. This enables the system to revisit predetermined locations for long-term studies, reducing its reliance on human supervision and maintenance and thereby reducing costs and increasing sampling rates. The system reliably captures data with a significantly higher temporal resolution than manual methods, regardless of atmospheric conditions. In addition, its modular design permits the incorporation of various sensor suites depending on the application and deployment site.

However, tethering a vehicle presents challenges both on land and in water because of the need to control the length of the cable being deployed (perhaps 200 m) and the magnitude of the pulling disturbance force exerted by the tether on the vehicle while encountering the vehicle's buoyancy and six degrees of freedom (DOFs) (Sattar et al., 2009). Some researchers use autonomous HUV (Aumeer et al., 2021) to contribute to the monitoring of Mauritius by collecting oceanographic data, surveyed images, and videos of coral reefs. The collected data were sent via Cellular IoT communication to a base station for analysis using two AI techniques: Machine Learning and Deep Learning. A comparison of the performance (Maslin et al., 2021) of semi-autonomous underwater vehicle (SAUV) inspections with diver-operated video recording (DOV) surveys for monitoring fish species on coral reefs.

#### 4.2.4. Swarm of robots

There are numerous examples of self-organizing colonies in the natural world, where large numbers of relatively simple agents interact locally to produce remarkable global behaviors. The result of this phenomenon is a system that is greater than the sum of its elements. Such recognized groups include insect colonies, bird flights, and fish schools (Kalantari et al., 2020).

Colonies of fish consisting of numerous individuals exhibit extraordinary behaviors such as long-distance migration, cohesive shoaling, and efficient food searching. To successfully avoid predators and capture prey, they exhibit a variety of morphologies and behaviors. A significant proportion of fish species exhibit schooling behavior, which provides advantages such as enhanced foraging, migration, and predator avoidance (Brönmark et al., 2014). Because fish make decisions based on visual observations or lateral line perceptions of nearby individuals, these collective behaviors emerge through implicit coordination. Through decentralization and reliance on local perception, fish schools successfully navigate underwater interactions, thereby achieving scalability and robustness (Berlinger et al., 2021a).

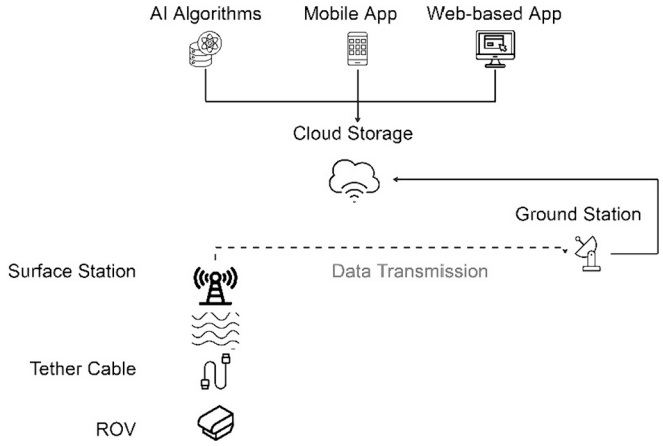


**Fig. 4.** Sample hybrid underwater vehicle, adapted from (Deng et al., 2014).

To better realize natural cooperative intelligence and create artificial robot cooperation, there has been a lot of effort to recognize the connection between local actions and global behaviors by trying to determine how individuals' behavior inside a group affects the group as a whole and how that group's actions, in turn, affect the individuals who make up that group. Recent advancements have demonstrated the viability of self-organized, homogeneous robot swarming inspired by nature (Ha and Tang, 2022). For example, autonomous systems, such as Intel's Shooting Stars and Crazyswarm, exist. A single computer that governs both systems relies on distinct technologies, such as GPS and motion capture, to (Preiss et al., 2017; Kushleyev et al., 2013). In contrast, VIO-Swarm determines its position using a local visual-inertial odometry (VIO) (Weinstein et al., 2018).

Comparatively, 3D marine autonomous systems have not yet achieved similar levels of self-organization to above-ground collectives. Several previous studies (Jaffe et al., 2017; Dunbabin et al., 2004; Zahadat and Schmickl, 2016; Schmickl et al., 2011; Katzschmann et al., 2018a; Yuh, 2000; Leonard et al., 2010) have suggested cooperative robots for applications spanning from monitoring the environment in ecologically fragile regions (such as coral reefs) to submarine equipment inspections and search-and-rescue missions. Furthermore, such collaborations can provide a synthetic method for comprehending how the decisions of individual fish robots lead to complexity in fish schools. However, underwater settings present significant challenges in terms of perception, transportation, communication, and sensing. Conventional aboveground communication methods, such as wireless radio, perform poorly underwater, and localization techniques, such as GPS, are inaccessible. Consequently, most underwater colonies coordinate only at the surface or not at all (Jaffe et al., 2017; Leonard et al., 2010).

The primary focus of underwater multi-robot systems has been on coordination accomplished through explicit and semi-centralized communication, in contrast to fish's implicit and perception-driven local coordination. However, this strategy has demonstrated limited efficacy. Comparing the extraordinary three-dimensional maneuverability of schooling fish or aerial robot swarms to the collaborative complexity of existing three-dimensional artificial underwater systems reveals a significant difference (Berlinger et al., 2021b). Inspired by fish schools that coordinate using vision, (Berlinger et al., 2021b) sought to create underwater robot collectives similar to fish schools, with high flexibility and autonomy from assistive technologies.

The spatial difference enables multiple perspectives necessary for undertaking high-resolution inspections and identifying and preventing obstacles. In addition, the distance between the nodes enables various sensing mechanisms to operate at optimal distances. Sonar nodes, for instance, are placed further away from objects than camera nodes. The variety of vehicles makes it possible to combine navigation and localization data; for instance, nodes sufficiently far away from obstructing structures may have more accurate navigation data than those closer to the structures. Moreover, its decentralized and physically disconnected nature permits a formation that can dynamically and instantly adapt to detect unanticipated environmental changes. This capability, called morphing, enables the formation to respond and adapt in real time to unpredicted situations (Kalwa et al., 2016).

#### 4.2.5. Cooperative multi-robots or autonomous sensors

A comprehensive survey of complex underwater structures typically requires the employment of a fleet of diverse cooperative vehicles or autonomous sensors (Kalwa et al., 2016). For example, a study (Shkurti et al., 2012) described a heterogeneous multi-robot system consisting of an aerial fixed-wing vehicle, an autonomous airboat, and a flexible-legged underwater robot. These robots communicate with remote specialists and function hierarchically. They identify areas of interest based on real-time aerial feedback and automatically collect visual data from various dimensions and underwater environments. Consequently, the system provides extensive aerial and underwater visual coverage of

designated reef locations. This method significantly improves monitoring efficiency by relieving scientists of tedious data collection duties and allowing them to concentrate on selecting survey regions.

A micro aerial robot, an autonomous surface vessel (ASV), and an autonomous underwater vehicle (AUV) to monitor algal outbreaks in coastal regions (Hodgson and Liebler, 2002). Some researchers (Mulhall, 2004) addressed the problem of coordinating multiple AUVs with a single UAV in a simulated grid world where the AUVs followed trajectories generated by the UAV. Researchers (Shihavuddin et al., 2013) created a collaborative system consisting of a torpedo AUV and an autonomous kayak for missions such as underwater mine clearance. The algorithm proposed (Beall et al., 2010) minimizes the positional error while an AUV traverses each sensor node, thereby facilitating data transport between a set of underwater sensor nodes. Using coordination between an underwater glider, surface drifter, and satellite information, scientists have demonstrated a second system for monitoring various oceanographic features. Additionally, (Grasmueck et al., 2006) devised a system that coordinated ASVs and UAVs using varying levels of autonomy, thereby facilitating data collection and remote monitoring by experts. Cost-effective visual data collection has been achieved by the use of floating cameras (Hoeke et al., 2009).

#### 4.2.6. Bio-inspired robots

A bio-inspired robot is a form of robot designed to mimic living organisms found in nature. These robots replicate various biological systems' structures, behaviors, and functionalities. A flexible underwater robot modeled after fish biology without tethers is ideally suited for navigating complicated underwater environments. A robot that can collect extensive three-dimensional sensor data can improve our knowledge of the ocean's dynamic microclimates while causing minimum disturbance to marine life (Hall et al., 2022). A flexible 3D-printed cable-driven flexible fish tail for usage on a soft fish robot was shown in (Hall et al., 2022) and its design and control settings were investigated to achieve maximum thrust force output.

The robotic fish is a unique biomimetic underwater vehicle with outstanding maneuverability and low noise. Because of these characteristics, it is useful for monitoring tasks, particularly for close-up observation. Tre has been a lot of research and development into building robotic fish that can move in various ways, such as cruise-straight swimming, cruise-in-turn diving, and ascent-decent gliding (Gao et al., 2021). Researchers created robots that can mimic the movements of fish and sharks and studied their swimming skills, propulsion mechanisms, and control methods (Gao et al., 2021). While most study has been conducted in the laboratory, some researchers are investigating the practical applications of robotic fish. These involve the development of gliding robotic fish for autonomously sampling water columns (Gao et al., 2021; Zhang and Negahdaripour, 2003) and developing biomimetic autonomous underwater vehicles for intelligence surveillance and reconnaissance (Gao et al., 2021). These studies tackle practical challenges, including watertight-cabin design, obstacle avoidance, and disturbance rejection.

### 4.3. Advantages and limitations of using robots for coral reef monitoring

Robots can offer a wide range of benefits for coral reef monitoring. Robots could be designed to work in a cluttered and hazardous environment and save many divers and scientists' lives. For instance, when monitoring reefs in areas where gill net fishing is common, including coastal coral communities in Southeast Asia, deploying ROV may be preferable to hiring scuba divers (Lam et al., 2007). Currently, biologists who "swim the reef" regularly to make observations are usually the ones that manually collect most visual measurements of marine life. This is risky, exhausting, and prone to mistakes. It should be possible to get more reliable data quickly, efficiently, and with lower risk by allowing the robot to perform some data-gathering tasks (Sattar et al., 2009). Also, applying robotics in surveying coral reef ecosystems increases the efficiency and accuracy of detection since temporal and spatial change information could be measured precisely by improving data rate collection and data quality by gathering data for a longer time period and in 3D dimension (Dunbabin et al., 2005). Visual 3D reconstruction algorithms give valuable ecological and habitat structure information from underwater photography.

On the other hand, an unaided swimmer or diver finds it challenging to navigate precisely across more expansive areas with the constant image overlap required for visual reconstruction. Underwater robots have demonstrated systematic coverage of regions much larger than the size of a single photograph, but access to appropriate robotic systems is restricted and necessitates skilled operators (Pizarro et al., 2017). There are fewer limitations on diving time and the geographical range of robots compared to human divers. The ROV (Lam et al., 2007) can capture images of transects across a larger geographical area without time constraints on diving. Importantly, robots can collect data and monitor 3D underwater environments without harming or causing damage to marine life and the undersea environment (Hall et al., 2022).

However, several difficulties are associated with using robots for coral reef monitoring. Typical above-ground communication methods like wireless radio have a weak signal underwater, and position localization techniques like GPS don't work (Berlinger et al., 2021b). One of the main technical issues is the communication problem. Multiple variables make teleoperation for undersea vehicles challenging. It isn't easy to communicate wirelessly via radio. Because salt water is conductive, where the beam's energy is quickly absorbed, conventional radio communications are impractical (Sattar et al., 2009). These transmissions are not really submerged. While radio waves are constrained by the thick underwater ecosystem and optical transmitters are affected by heat and photons, acoustic modems, which are incredibly expensive, are the only remaining choice (Avila et al., 2021b). Since acoustic modems need much power to transmit over long distances and have severe bandwidth restrictions (between 1 and 10 kbps), they are currently used in marine systems. Real-time video transmission is not possible due to the limited bandwidth. Even human scuba divers frequently use short-range visual communications like rudimentary sign language for underwater task coordination. For this reason, underwater vehicle operators are advised to those visual signals (Sattar et al., 2009).

Another key challenge in using robots is navigation in free space (Girdhar and Dudek, 2016). Frequently, regions with a large diversity of reefs also happen to be those areas that are geologically or ecologically interesting; examples include cold-water coral reefs, the Mid-Atlantic Ridge volcanoes, and even artificial structures like wind turbine bases. The current techniques for navigation, such as side-scan sonar and static formation, are constrained in these regions with very irregular surfaces. Similarly, today's autonomous vehicles are frequently inappropriate for functioning in these difficult, unstructured situations. Other than self-navigation, other autonomy issues are challenging in these conditions, including online mapping and sensing and real-time path planning in an incomplete environment. There is a genuine risk of vehicle damage or loss in addition to solving these high-level autonomy challenges in a sparsely populated area is commonly anticipated with AUV operations (Kalwa et al., 2016). Caves, overhangs, hard and soft corals, boulders, powerful currents, varied depths, and marine life all contribute to the underwater environment's difficulty in navigating. All of these elements combine to make navigation tasks more difficult by creating a chaotic and non-planar environment. Extensive maneuvering and navigational skills are needed to gather accurate information in such a setting (Maslin et al., 2021). Even under the greatest circumstances, performing accurate localization and navigation in highly unstructured underwater coral reef habitats is challenging. Underwater research and commercial vehicles typically employ pricey acoustic location and sonar equipment with a sizable external infrastructure to function properly. Several studies, including (Dunbabin et al., 2006), concentrated on building an accurate vision-based motion estimation method with low-priced sensors for real-time, untethered environmental monitoring in the Great

Barrier Reef. The goal is to do so without using acoustic locating technology. Using this method, as shown by experiments, it is possible to get precise odometry and terrain profile data that may be used as input for the vehicle's operations.

For reef surveying applications, it is essential to identify the position and distance traveled by an AUV to obtain accurate and reproducible data. It is essential to have precise odometry to guarantee that survey paths are followed precisely. Several strategies are used to estimate the motion of a vehicle. Doppler velocity logs and other acoustic sensors are frequent methods for acquiring precise motion data. To facilitate navigation, station maintenance, and give feedback information for manipulator duties, vision-based motion estimation is becoming increasingly popular in underwater applications (Van Der Zwaan et al., 2002; Santos-Victor and Sandini, 1997). However, factors like visibility, illumination, and optical distortion due to differences in refractive indices all affect the precision with which one can see underwater. Corrective lenses or careful calibration methods are often necessary to overcome these obstacles (Bryant et al., 2000; Dunbabin et al., 2006). By employing an inverse sensor model for range sensors, it is possible to calculate real-world locations that maximize the utility of sensor readings in determining the precise position and shape of obstacles. (Grabowski et al., 2003) proposed an exploration strategy centered on maximizing the comprehension of obstacles instead of merely investigating empty spaces. In this method, the robot identifies the next-best view area in which obtaining a sonar sensor reading would substantially enhance the quality of the obstacle representation (Girdhar and Dudek, 2016). Without an external localizer, it is advantageous for the robot to simultaneously investigate, map, and localize itself in the environment. In contrast to terrestrial environments, obstacle avoidance, and map construction are typically not the primary focus in aquatic and aerial environments (Girdhar and Dudek, 2016).

### 4.4. Artificial intelligence, machine learning and robotics in coral reef monitoring

To effectively address the potential collapse of diverse ecosystems, it is essential to have accurate and reliable ecological data, especially in a rapidly changing world. Hypothesis-driven and adaptive management approaches are crucial for effective monitoring and conservation efforts. However, our ability to collect and analyze ecological data has been limited, necessitating more effective ecological surveys. Therefore, It is of the utmost importance to reduce the time and cost of conducting such surveys (González-Rivero et al., 2020). As long as field data is still needed, the main challenge is shifting from generating information to making sure that new tools can turn it into meaningful metrics that can help us learn more about the effects of a fast-changing environment on coral reefs (González-Rivero et al., 2016). The task of quickly and accurately analyzing many images has led marine science and computer science to work together in a productive manner. Even though satellite image analysis (Mountrakis et al., 2011) and plankton ecology (Culverhouse et al., 2006) use automatic image analysis a lot, it is still pretty new concerning coral reef systems. These technologies, which frequently apply machine learning to map the visual characteristics of photographs to their semantic classifications, enable marine biologists to gather valuable ecological data from photographic recordings (Beijbom et al., 2015), considerably faster than by hand (González-Rivero et al., 2014). Also, the advancement in imaging sensors and computer vision approaches has made it possible to use new methods in coral reef ecology to measure a variety of other parameters that are important to the field, such as the complexity of reef terrain (Friedman et al., 2010; Leon et al., 2015) and the number of fish (Boom et al., 2014).

Integrating artificial intelligence (AI) and machine learning offers a promising solution for coral reef monitoring's data processing bottleneck (González-Rivero et al., 2020). AI and machine learning techniques have a wide variety of applications in coral reef monitoring, such as recognition and classification of coral reef species using identification and tracking, data fusion to combine and analyze the data coming from multiple sources and sensors, Coral reef counting, and building a predictive model for forecasting future changes.

Numerous studies exist on coral categorization (Mahmood et al., 2016; Beijbom et al., 2015), detection (Mahmood et al., 2016; Beijbom et al., 2015), and classification (Manderson et al., 2017b; Mahmood et al., 2016; Beijbom et al., 2015). A CNN detector and tracking for coral reef species was developed in (Modasshir et al., 2021) and (Modasshir et al., 2018a). (IEEE Robotics and Automation Society and Institute of Electrical and Electronics Engineers, 2018) redesigned the RetinaNet (Lin et al., 2020) to train a detector for eight coral species. Among numerous vision-based trackers (Li et al., 2018), the online performance of the Kernelized Correlation Filters (KCF) (Henriques et al., 2015) in AUVs is acceptable. During real-time monitoring, the KCF tracking method can determine when an object leaves the field of view.

The integration of robotics with AI and ML is improving the potential of coral reef monitoring; This combination has transformed how ecological surveys are being conducted, particularly in challenging environments, by enabling the development of efficient and integrated reef monitoring system (Mellin et al., 2020). Additionally, the incorporation of AI tools, such as machine learning, has shown promise in increasing the efficiency of reef monitoring, therefore, enhancing our ability to assess the future of coral reefs in the context of rapid environmental change (de Pablo et al., 2021). One example of this integration is the use of swarm algorithms inspired by natural behaviors, which are forms of artificial intelligence; These algorithms, such as Particle Swarm Optimization and Artificial Bee Colony, are adapted for underwater conditions, allowing for optimized robotic movements and decision-making processes; contributing to advanced monitoring when enabling efficient navigation and data collection (Connor et al., 2021). Another example is the use of machine learning for mapping visual characteristics to semantic classifications, which aids in accurate ecological data collection from photographic recordings (Christensen et al., 2022). A clear positive outcome of this use of AI in underwater robotics is improved navigation skills in complex underwater environments, accuracy in data collection, and data analysis contributing to a successful and more efficient marine research and conservation efforts.

#### 4.4.1. Recognition and classification of coral reef species

Advancements in automatic image recognition and classification could expedite coral reef monitoring image analysis (González-Rivero et al., 2020). Scientists can now capture and systematically explain large-scale ecological processes with genus-specific resolution using scalable technologies such as photo-mosaics (Finney and Stephen, 2005; González-Rivero et al., 2014). (Beijbom et al., 2012) looked at how autonomous approaches can be used to use survey pictures to predict the spatial distribution of different organisms in coral reef ecosystems. Their work used support vector machine algorithms to classify images by extracting image patches around sparse labels. Machine learning methods, like k-nearest neighbors, have also been used in other studies to look at coral reefs and what they are like (Manderson et al., 2017b; Mary and Dharma, 2017; Shihavuddin et al., 2013). Still, deep learning systems have made huge steps forward in tasks like classification, detection, and segmentation (Garcia-Garcia et al., 2017). Scientists (Moniruzzaman et al., 2017) showed that these changes have also been made in studying coral reefs.

Deep learning has raised the standard for image recognition (González-Rivero et al., 2020). Deep learning methods also work better when they can learn from many different kinds of data, which has also been shown (Beijbom et al., 2016). For example, (Zweifler et al., 2017) has made a fluorescence imaging system called FluorIS with a wide field of view that can classify coral species more correctly than RGB images alone. Classification is being replaced by semantic segmentation, which can provide more information (pixel level) than classification alone. The initial methods classified picture patches to rebuild the image segmentation (Redmon et al., 2016). Nevertheless, around the margins of

segmented regions, the precision of these patch-based methods is often limited. In addition, they must be executed the same number of times as the number of picture patches cropped to obtain the fully segmented image.

Deep learning has impacted object detection systems, leading to standard vision guidelines in cutting-edge object detectors. These object detection algorithms have been shown to be effective in various domains. Being point annotated, the majority of coral detection datasets (Beijbom et al., 2012) and (Beijbom et al., 2015) are incompatible with object detection algorithms that require bounding box annotation.

A new way to improve the detection model through self-training is presented (Modasshir and Rekleitis, 2020), using accurate predictions of coral objects or tracked instances. Self-training means that a trained model can use its own results to train itself even more. These corals that have been found and are being watched are called "soft-labels," and they are used to add to the training data for the recognition model. But including all soft labels in training might not be a good idea because the detecting model might lose track of some things and make some wrong predictions. To deal with this, a classification network is used to weed out "harmful" soft labels and make sure they aren't used too much during training. The self-training approach starts by training the model on fully labeled data. The model is then retrained using pseudo-labels made from partially labeled or unlabeled data. This process is done over and over again (Muhlenbach and Zighed, 2004). Also, adding hard examples, called "difficult examples mining," has been shown to help make object trackers work better (Shrivastava et al., 2016).

Traditional or surface-level machine learning techniques, such as Support Vector Machine, have limitations because they depend on manually designed feature extractors to transform raw image data (pixel values) into representations suitable for the learning algorithm to classify objects within an image. Nevertheless, one of the remarkable developments of deep learning is its ability to autonomously learn the necessary features for classification, allowing it to manage complex structures in high-dimensional data effectively.

#### 4.4.2. Data fusion of multiple sensors

A data fusion technique (Dunbabin et al., 2005) that combines vision and inertial sensors can accomplish better performance comparable to acoustic sensor systems while significantly reducing the cost of the vehicle. The system described in this paper is intended to address the need for a low-cost vehicle capable of operating in highly unstructured environments without requiring extensive human intervention or external positioning infrastructure. By intelligently integrating low-resolution sensors and hardware, the system provides accurate localization and navigation data estimates. In underwater navigation, it is common to combine visual and acoustic information to improve sensor resolution and accuracy (Eustice et al., 2004). Even though this fusion method can produce more accurate motion estimation than vision alone, it is typically implemented offline and is primarily utilized in scenarios involving deeper water.

#### 4.4.3. Counting coral reefs

Counting objects is a hotly debatable area of study in AI, and counting the coral density is a challenging problem for several reasons, such as underwater visibility, color suppression, and a hazing environment. Objects in the field of view are frequently so unclear that neither humans nor deep models can identify them. In addition, coral counting involves both temporal and spatial dimensions. It is crucial to avoid following image frames from recalculating after an object has been recognized. The detection process frequency and model must be carefully considered when attempting detection and counting with a low-powered AUV.

Many algorithms have been established to deal with the problem of counting in the visual domain. Most of these algorithms are either based on regression or detection. Regression-based techniques Using CNN-based models, (Boominathan et al., 2016; Pham et al., 2015; Xu and Qiu, 2016; Sam et al., 2017; Zhang et al., 2016) estimate the number of objects in an image without explicitly classifying or localizing them. In contrast, detection-based approaches (Redmon et al., 2016; Gao et al., 2016) use a detector, frequently based on convolutional neural networks (CNNs), to locate objects in an image by enclosing them with bounding boxes. The number of these bounding frames is then utilized to estimate the number of objects in the image.

To collect visual data, marine scientists execute surveys across various sections of the coral reef, following transects such as straight lines or rectangular paths. Experts then manually analyze the recorded videos to enumerate or annotate coral species to determine population density. An automated method for accomplishing this mission is also presented (Modasshir et al., 2018b). The research modifies the RetinaNet deep learning network to accommodate the coral dataset's limited training examples and imbalanced class distribution. The augmented network is able to identify and locate various coral species within images. The system operates by continually receiving photographs as input from either an autonomous underwater vehicle (AUV) or a hand-held camera. These images are then processed by a Convolutional Neural Network (CNN), which enables the identification and localization of diverse coral species by means of bounding frames. Additionally, the system employs OpenCV2's KCF tracker (Henriques et al., 2015) to monitor detected bounding boxes across successive images. If a new detection occurs near a tracked bounding box, the existing bounding box's coordinates are updated without the coral count being incremented. This offline-capable method streamlines the identification process by automating the identification and enumeration of coral reefs.

## 5. Case studies of robotics in coral reef monitoring

### 5.1. Examples of successful applications of robotics in coral reef monitoring

Numerous remotely operated vehicles (ROVs) and autonomous underwater vehicles (AUVs) are deployed globally for a variety of monitoring applications (Whitcomb et al., 2000). However, these vehicles are typically large, expensive, reliant on an extensive external infrastructure for precise positioning, and require multiple operators. Furthermore, these vehicles tend to avoid extremely unstructured reef environments, such as Australia's Great Barrier Reef. Comparatively, in regions with less complex reef structures, the challenges might differ, focusing more on technological capabilities and environmental factors. This global diversity in applications highlights the adaptability and varying effectiveness of robotic interventions in different coral reef conditions (Arain et al., 2020). Limited research has been conducted on shallow water applications and reef traversal with these vehicles. ROVs have been used for video transects, and the identification of biomass in deeper areas, but human operators are still necessary for the process (Dunbabin et al., 2006).

As part of Australia's Integrated Marine Observation System (IMOS), the Australian Centre for Field Robotics employs Sirius, an ocean-going Autonomous Underwater Vehicle (AUV). Sirius conducts georeferenced, high-resolution surveys at various locations along the Australian coastline, facilitating long-term monitoring through repeated surveys. In contrast to selecting diverse sites within a region, precisely revisiting the exact benthos area over multiple years provides greater statistical power to detect temporal changes (Nguyen et al., 2020). This method also facilitates investigating changes in individual organisms, such as a single coral polyp, providing marine biologists and ecologists with new opportunities. Sirius uses various navigation sensors, including Global Positioning Systems (GPS) and Ultra-Short BaseLine (USBL) positioning, to guarantee accurate multi-year change detection, achieving a localization accuracy of approximately 5 m underwater. However, documenting sub-centimeter-scale changes that can occur over multiple years in habitats such as coral reefs requires even more precise registration. Therefore, improved registration capabilities are required for

reliable multi-year change detection.

Moreover, an underwater robot was designed specifically for monitoring coral reefs in Honduras, in this case study, the purpose was to simplify challenges related to underwater mechanical robot design and to integrate a vision system for capturing videos and photos for monitoring purposes. The main challenges included difficulties related to mechanical design and ensuring effective internal monitoring for leaks and temperature. To address these challenges, the team implemented the VDI 2206 model, involving lab design and prototype integration phases. They conducted various force estimation simulations and field tests to ensure the robot could accomplish its tasks, with special focus on PWM outputs for controlling thrust and internal sensors for ensuring operational integrity (Avila et al., 2021a).

### 5.2. Review of studies that have used robots to address specific research questions in coral reef ecology

Robots have previously been used in underwater surveillance in coral reef research, and they have proven to be useful tools for studying and protecting these ecosystems (Loya et al., 2019). These modern robotics technologies have the potential to greatly enhance our understanding of marine ecosystems and advance coral reef conservation efforts (Ferrera et al., 2021). Robotics applications in coral reef research show significant geographical diversity. For instance, conducted a study on the biogeography and molecular diversity of coral symbionts in the Arabian Peninsula, highlighting differences in symbiont taxon richness over geographical regions, demonstrating the geographical diversity of coral reef research (Ziegler et al., 2017). In some regions, robots equipped with environmental sensors have been key in monitoring changes (Hedley et al., 2016b).

A previous study reported that an underwater robot was developed, which was equipped with sensors to monitor coral reefs for temperature changes (Muller-Karger et al., 2018), salinity, dissolved oxygen, pressure, and pH levels (Sánchez-Ferreira et al., 2016) This underwater robot even can collect water samples for additional laboratory testing. This can be a valuable tool for monitoring water quality and ensuring effective conservation efforts (Katzschmann et al., 2018b).

Overall these robots can collect data in real-time and provide a more accurate and comprehensive picture of coral reef health; they can also be used in seafloor exploration and surveillance of divers (Muller-Karger et al., 2018; Sánchez-Ferreira et al., 2016; Bicknell et al., 2016). Robots have been used to mimic the behavior of predator fish to study the interactions between predators and prey. These robots were programmed to behave in specific ways, test different scenarios, and observe the responses of prey species. It has been reported that prey fish species' fear and stress response are enhanced after exposure to a robotic predator (Woodhead et al., 2019). Robots have been used to collect direct coral samples to analyze potential threats (González-Sabbagh and Robles-Kelly, 2023). This can be done more efficiently and safely by using robotic grippers (UK-RAS Network Robotics and Autonomous Systems, 2018) as an alternative to diving to collect samples. Researchers have developed underwater robots that can plant coral fragments onto damaged reefs. These robots can work faster and more efficiently than human divers, allowing for larger-scale restoration efforts (Wilson et al., 2015). Altogether, robots can potentially revolutionize how to protect and monitor coral reefs. However, more research is needed to fully understand the capabilities and limitations of robotics technologies.

### 5.3. Discussion of the implications of these studies for future research and conservation efforts

Robot technology-based coral reef studies can significantly impact future research and conservation measures. Coral reefs are extremely complex ecosystems, and studying them can be challenging due to their size, depth, and location (Murad et al., 2016). Robots, however, can overcome the coral reef diversity challenges by allowing marine conservationists to explore more in-depth and study these environments more in detail (Wang et al., 2022; Zantis et al., 2021; Farid et al., 2006). Robotics in coral reef research and conservation can also help raise local awareness of the importance of these marine ecosystems and potential threats to them. By making research more accessible and engaging, robots can help inspire greater public support for conservation measures (Farid et al., 2006; Sahoo et al., 2019). Moreover, robotic technologies have the potential to increase the efficiency and accuracy of coral reef research significantly (Muller-Karger et al., 2018; Luong et al., 2019).

Robots can collect data more quickly and precisely, leading to a more comprehensive understanding of reef health (Avila et al., 2021a). This can lead to more targeted and effective conservation efforts. Robots can access difficult and dangerous premises, particularly benthic marine areas, more quickly and safely, providing valuable information on previously inaccessible areas of reef (Voolstra, 2020). Additionally, robotic technologies can potentially reduce the cost of coral monitoring and make it more accessible for conservationists (Girdhar et al., 2019; UK-RAS Network Robotics and Autonomous Systems, 2018). Therefore, robot technology in coral reef studies has significant implications for future research and conservation measures. With the data gathered from these technologies, conservationists can develop more effective measures to protect and preserve these unique and vital reef ecosystems.

## 6. Discussion and future direction

Based on the analysis of research gaps and trends in the field, several areas require further research in monitoring coral reefs, including coral nurseries and the open sea. Coral nurseries are artificial structures that are considered to cultivate and propagate corals (Gao et al., 2019). They are typically made of materials like stainless steel, PVC pipes, ropes, and nets and are placed in shallow, nutrient-rich waters where corals can propagate and bloom. These nurseries help restore damaged or degraded coral reefs, which are important ecosystems that shelter a wide range of marine life (Fayemi et al., 2017).

By growing new corals in nurseries and then transplanting them onto damaged reefs, conservationists hope to help these ecosystems recover and become more resilient to future threats like climate change (Knowlton and Jackson, 2008), pollution (Rädecker et al., 2021), and overfishing (Fine et al., 2005). Growing corals in nurseries typically involves collecting small fragments of healthy corals, which are then attached to the nursery structures and monitored for growth and health (Gao et al., 2019). Once the corals have reached a certain size and are deemed healthy enough, they can be transplanted onto damaged reefs. Coral nurseries are often combined with other conservation strategies, such as reducing pollution and protecting reefs from overfishing and destructive practices. They are an important tool for helping to preserve and restore these important marine ecosystems.

One crucial area is the development of robotic systems for monitoring these coral nurseries. Robots can play a significant role in coral nurseries by providing more efficient and effective ways to maintain and propagate. These robotic systems could be designed to navigate shallow waters, detect damage, recruit corals and larvae, and collect data on young corals' growth and survival rates. Additionally, these systems can be programmed to accurately and efficiently fragment corals into smaller pieces, which can then be grown into larger colonies in a controlled environment. Another important area is the integration of sensors and imaging systems into robotics for improving the data collection in nurseries to monitor their health and condition. For instance, cameras and hyperspectral sensors could be used to detect changes in water quality, temperature, and salinity, which can significantly impact coral growth. These systems can also be used to transplant healthy corals onto damaged reef areas. Robots can do the transplantation process more precisely and with less far damage to the coral. Moreover, robots have great potential to maintain coral nurseries by eliminating any debris that may accumulate on the structures.

In addition to the application of robotic systems for monitoring coral

nurseries, several other key areas demand additional investigation in coral reef monitoring. One example is monitoring coral reefs in indoor spaces such as aquariums and laboratories. Coral reefs cultivated indoors can be useful research instruments for studying coral biology, ecology, and conservation. However, these settings can be difficult to maintain because they require careful management of environmental variables such as temperature, lighting, water quality, and nutrient levels. These environmental variables can be automated and optimized using robotic devices, lowering the danger of coral stress and disease outbreaks. For instance, robots can be programmed to control water temperature, lighting, and nutrient levels in an aquarium or laboratory. This automation can help to ensure that the coral reef environment is kept in optimum condition, which can benefit the coral's overall health and well-being. Furthermore, robotic systems can be used to monitor coral reefs in interior spaces, gathering data on environmental variables and informing operators of any changes or trends in the coral's health.

Robotic systems can also be utilized in outside locations, such as the open sea, to monitor environmental variables and optimize coral reef health. These systems can be programmed to operate autonomously and investigate coral reef environments while gathering data on temperature, salinity, nutrient levels, and other parameters affecting coral growth and health. These systems' data can be utilized to optimize the placement of coral nurseries and indicate sections of the reef that require additional conservation efforts. Monitoring coral reefs in the open sea provides particular problems since these areas are frequently exposed to unexpected weather and a variety of potential stressors, such as ocean acidification and rising sea temperatures. Robotic systems can help address some of these issues by offering a more consistent and comprehensive coral reef monitoring platform. Lastly, given the differences between robotic and traditional coral reef monitoring approaches, a hybrid approach emerges as an opportunity for future research. The combination of robotic technology and traditional monitoring approaches may take advantage of the strengths of each, overcoming the limitations associated with each method when used separately.

## 7. Conclusion

Coral reefs are essential ecosystems that provide critical services to humans and the marine environment. However, climate change, pollution, overfishing, and other human activities represent substantial hazards to coral reefs. Monitoring and conservation activities are critical for preserving these important ecosystems. The employment of robotic systems to monitor coral reefs, both indoors and outdoors, has enormous potential for boosting conservation efforts. Robotic systems can aid in the automation and optimization of environmental variables, the detection of damage and changes in water quality, and the precise and efficient propagation and transplantation of corals. Furthermore, the information obtained by these systems can be used to improve conservation methods and provide crucial insights into the health and condition of coral reefs. Moreover, is important to understand that while robotic systems have remarkable capabilities, they should be considered complementary, rather than replacements for existing conservation methods. As a result, it is critical to continue studying in this area and fully exploit the potential of robotic systems in monitoring and maintaining coral reefs for future generations.

## CRediT authorship contribution statement

**Jennifer A. Cardenas:** Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Methodology, Data curation, Conceptualization. **Zahra Samadikhoshkho:** Writing – review & editing, Writing – original draft, Validation, Data curation. **Ateeq Ur Rehman:** Writing – review & editing, Writing – original draft, Visualization, Validation, Data curation. **Alexander U. Valle-Pérez:** Writing – original draft, Visualization, Validation, Data curation. **Elena Herrera-Ponce de León:** Writing – original draft, Visualization, Validation. **Charlotte A.E. Hauser:** Writing – review & editing, Validation, Supervision, Methodology, Investigation. **Eric M. Feron:** Writing – review & editing, Validation, Supervision. **Rafiq Ahmad:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. NSERC funding: NSERC AOO 586056-23 Ahmad & NSERC RGPIN-2017-04516 Ahmad.

## Data availability

No data was used for the research described in the article.

## References

Albalawi, H.I., et al., 2021. Sustainable and eco-friendly coral restoration through 3D printing and fabrication. ACS Sustain. Chem. Eng. https://doi.org/10.1021/acssuschemeng.1c04148.

Alonso, I., Yuval, M., Eyal, G., Treibitz, T., Murillo, A.C., Dec. 2019. CoralSeg: learning coral segmentation from sparse annotations. J Field Robot 36 (8), 1456–1477. https://doi.org/10.1002/rob.21915.

Arain, B., Dayoub, F., Rigby, P., Dunbabin, M., Jan. 2020. Close-Proximity Underwater Terrain Mapping Using Learning-based Coarse Range Estimation [Online]. Available: http://arxiv.org/abs/2001.00330.

Aumeer, W., Pooloo, N., Khoodeeram, R., 2021. A proposed iot architecture for corals research using AI and robotics. In: Advances in Intelligent Systems and Computing. Springer Science and Business Media Deutschland GmbH, pp. 371–385. https://doi.org/10.1007/978-981-33-4299-6_31.

Avila, J.L.O., Avila, M.G.O., Perdomo, M.E., 2021a. Design of an underwater robot for coral reef monitoring in Honduras. In: 2021 6th International Conference on Control and Robotics Engineering, ICCRE 2021. Institute of Electrical and Electronics Engineers Inc., pp. 86–90. https://doi.org/10.1109/ICCRE51898.2021.9435710. Apr.

Avila, J.L.O., Avila, M.G.O., Perdomo, M.E., 2021b. Design of an underwater robot for coral reef monitoring in Honduras. In: 2021 6th International Conference on Control and Robotics Engineering, ICCRE 2021. Institute of Electrical and Electronics Engineers Inc., pp. 86–90. https://doi.org/10.1109/ICCRE51898.2021.9435710. Apr.

Avila-Ramírez, A., et al., 2021. Ecologically friendly biofunctional ink for reconstruction of rigid living systems under wet conditions. Int. J. Bioprint. 7 (4), 1–11. https://doi.org/10.18063/ijb.v7i4.398.

Baker, A.C., Glynn, P.W., Riegl, B., Dec. 2008. Climate change and coral reef bleaching: an ecological assessment of long-term impacts, recovery trends and future outlook. Estuar. Coast. Shelf Sci. 80 (4), 435–471. https://doi.org/10.1016/J.ECSS.2008.09.003.

Beall, C., Lawrence, B.J., Ila, V., Dellaert, F., 2010. 3D reconstruction of underwater structures. In: IEEE/RSJ 2010 International Conference on Intelligent Robots and Systems, IROS 2010 - Conference Proceedings, pp. 4418–4423. https://doi.org/10.1109/IROS.2010.5649213.

Behmanesh, M., Hong, T.S., Kassim, M.S.M., Azim, A., Dashtizadeh, Z., May 2017. A brief survey on agricultural robots. Int. J. Mech. Eng. Robot. Res. 6 (3), 178–182. https://doi.org/10.18178/ijmerr.6.3.178-182.

Beijbom, O., Edmunds, P.J., Kline, D.I., Mitchell, B.G., Kriegman, D., 2012. Automated annotation of coral reef survey images. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition, pp. 1170–1177. https://doi.org/10.1109/CVPR.2012.6247798.

Beijbom, O., et al., Jul. 2015. Towards automated annotation of benthic survey images: variability of human experts and operational modes of automation. PloS One 10 (7). https://doi.org/10.1371/journal.pone.0130312.

Beijbom, O., et al., 2016. Improving automated annotation of benthic survey images using wide-band fluorescence. In: Scientific Reports, 6. Nature Publishing Group. https://doi.org/10.1038/srep23166. Mar. 29.

Berkelmans, R., Van Oppen, M.J.H., Sep. 2006. The role of zooxanthellae in the thermal tolerance of corals: a 'nugget of hope' for coral reefs in an era of climate change. Proc. R. Soc. B Biol. Sci. 273 (1599), 2305–2312. https://doi.org/10.1098/rspb.2006.3567.

Berlian, M.H., et al., Feb. 2017. Design and implementation of smart environment monitoring and analytics in real-time system framework based on internet of underwater things and big data. In: Proceedings - 2016 International Electronics Symposium, IES 2016. Institute of Electrical and Electronics Engineers Inc., pp. 403–408. https://doi.org/10.1109/ELECSYM.2016.7861040

Berlinger, F., Gauci, M., Nagpal, R., 2021a. Implicit Coordination for 3D Underwater Collective Behaviors in a Fish-inspired Robot Swarm [Online]. Available: https://www.science.org.

Berlinger, F., Gauci, M., Nagpal, R., 2021b. Implicit coordination for 3D underwater collective behaviors in a fish-inspired robot swarm. Sci. Robot. 6 (50) https://doi.org/10.1126/SCIROBOTICS.ABD8668.

Bicknell, A.W.J., Godley, B.J., Sheehan, E.V., Votier, S.C., Witt, M.J., 2016. Camera technology for monitoring marine biodiversity and human impact. Front. Ecol. Environ. 14 (8), 424–432. Wiley Blackwell. Oct. 01. https://doi.org/10.1002/fee.1322.

Biswas, N., Maurya, P., Navelkar, G., 2021. Navigation system for a coral reef monitoring robot (C-bot). In: Oceans Conference Record (IEEE). Institute of Electrical and Electronics Engineers Inc. https://doi.org/10.23919/OCEANS44145.2021.9705959

Boom, B.J., et al., 2014. A research tool for long-term and continuous analysis of fish assemblage in coral-reefs using underwater camera footage. Ecol. Inform. 23, 83–97. https://doi.org/10.1016/j.ecoinf.2013.10.006.

Boominathan, L., Kruthiventi, S.S.S., Babu, R.V., Aug. 2016. CrowdNet: A Deep Convolutional Network for Dense Crowd Counting [Online]. Available: http://arxiv.org/abs/1608.06197.

Brodie, J., Fabricius, K., De'ath, G., Okaji, K., 2005. Are increased nutrient inputs responsible for more outbreaks of crown-of-thorns starfish? An appraisal of the evidence. Mar. Pollut. Bull. 266–278. https://doi.org/10.1016/j.marpolbul.2004.10.035.

Brönmark, C., et al., 2014. There and back again: Migration in freshwater fishes. Can. J. Zool. 92 (6), 467–479. https://doi.org/10.1139/cjz-2012-0277. National Research Council of Canada.

Bryant, M., Wettergreen, D., Abdallah, S., Zelinsky, A., 2000. Robust camera calibration for an autonomous underwater vehicle [online]. Available: http://wwwsyseng.anu.edu.au/rsl/.

Bryant, D.E.P., et al., Oct. 2017. Comparison of two photographic methodologies for collecting and analyzing the condition of coral reef ecosystems. Ecosphere 8 (10). https://doi.org/10.1002/ecs2.1971.

Cambria, E., White, B., 2014. Jumping NLP curves: a review of natural language processing research. IEEE Comput. Intelligence Mag. 9 (2), 48–57. https://doi.org/10.1109/MCI.2014.2307227. Institute of Electrical and Electronics Engineers Inc.

Capocci, R., Dooly, G., Omerdić, E., Coleman, J., Newe, T., Toal, D., 2017. Inspection-class remotely operated vehicles-a review. J. Mar. Sci. Eng. 5 (1) https://doi.org/10.3390/jmse5010013. MDPI AG, Mar. 01.

Carleton, J.H., Done, T.J., 1995. Quantitative Video Sampling of Coral Reef Benthos: Large-scale Application. https://doi.org/10.1007/BF00304070.

Chennu, A., Färber, P., De'ath, G., De Beer, D., Fabricius, K.E., Dec. 2017. A diver-operated hyperspectral imaging and topographic surveying system for automated mapping of benthic habitats. Sci. Rep. 7 (1) https://doi.org/10.1038/s41598-017-07337-y.

Christensen, L., de Gea Fernández, J., Hildebrandt, M., Koch, C.E.S., Wehbe, B., Aug. 2022. Recent advances in AI for navigation and control of underwater robots. Curr. Robot. Rep. 3 (4), 165–175. https://doi.org/10.1007/s43154-022-00088-3.

Connor, J., Champion, B., Joordens, M.A., 2021. Current algorithms, communication methods and designs for underwater swarm robotics: a review. IEEE Sensors J. 21 (1), 153–169. Institute of Electrical and Electronics Engineers Inc.. Jan. 01. https://doi.org/10.1109/JSEN.2020.3013265.

Côté, I.M., Gill, J.A., Gardner, T.A., Watkinson, A.R., Feb. 2005. Measuring coral reef decline through meta-analyses. In: Philosophical Transactions of the Royal Society B: Biological Sciences. Royal Society, pp. 385–395. https://doi.org/10.1098/rstb.2004.1591.

Culverhouse, P.F., Williams, R., Benfield, M.C., Flood, P.R., Sell, A.F., Mazzocchi, M.G., Buttino, I., & Sieracki, M.E. (2006). Automatic image analysis of plankton: future perspectives. Mar. Ecol. Prog. Ser., 312, 297-309.

de Pablo, L.X., et al., Dec. 2021. A doubling of stony coral cover on shallow forereefs at Carrie Bow Cay, Belize from 2014 to 2019. Sci. Rep. 11 (1) https://doi.org/10.1038/s41598-021-96799-2.

De'Ath, G., Fabricius, K.E., Sweatman, H., Puotinen, M., Oct. 2012. The 27-year decline of coral cover on the Great Barrier Reef and its causes. Proc. Natl. Acad. Sci. U. S. A. 109 (44), 17995–17999. https://doi.org/10.1073/pnas.1208909109.

Deng, Z., Zhu, D., Xu, P., Fang, J., 2014. Hybrid Underwater Vehicle: ARV Design and Development [Online]. Available: http://www.sensorsportal.com.

Dimoff, S.A., Halliday, W.D., Pine, M.K., Tietjen, K.L., Juanes, F., Baum, J.K., May 2021. The utility of different acoustic indicators to describe biological sounds of a coral reef soundscape. Ecol. Indic. 124, 107435 https://doi.org/10.1016/J.ECOLIND.2021.107435.

Ditria, E.M., Buelow, C.A., Gonzalez-Rivero, M., Connolly, R.M., 2022. Artificial intelligence and automated monitoring for assisting conservation of marine ecosystems: a perspective. Front. Mar. Sci. 9 https://doi.org/10.3389/fmars.2022.918104. Frontiers Media S.A., Jul. 28.

Dunbabin, M., Roberts, J., Usher, K., Corke, P., 2004. A New Robot for Environmental Monitoring on the Great Barrier.

Dunbabin, M., Roberts, J., Usher, K., Winstanley, G., Corke, P., 2005. A hybrid AUV design for shallow water reef navigation. In: Proceedings - IEEE International Conference on Robotics and Automation, pp. 2105–2110. https://doi.org/10.1109/ROBOT.2005.1570424.

Dunbabin, M., Usher, K., Corke, P., 2006. Visual Motion Estimation for an Autonomous Underwater Reef Monitoring Robot, 25. https://doi.org/10.1007/978-3-540-33453-8_4.

Duong, L.N.K., et al., 2020. A review of robotics and autonomous systems in the food industry: from the supply chains perspective. In: Trends in Food Science and Technology, 106. Elsevier Ltd, pp. 355–364. Dec. 01. https://doi.org/10.1016/j.tifs.2020.10.028.

Edmunds, P.J., et al., 2018. Critical information gaps impeding understanding of the role of larval connectivity among coral reef islands in an era of Global Change. Frontiers in Marine Science 5 (AUG). https://doi.org/10.3389/fmars.2018.00290. Frontiers Media S.A., Aug. 27.

Eleftheriou, A., 2013. Methods for the Study of Marine Benthos. Wiley-Blackwell.

Eustice, R., Pizarro, O., Singh, H., 2004. Visually augmented navigation in an unstructured environment using a delayed state history. In: Proceedings - IEEE International Conference on Robotics and Automation. Institute of Electrical and Electronics Engineers Inc., pp. 25–32. https://doi.org/10.1109/robot.2004.1307124

Fabricius, K.E., Okaji, K., De'ath, G., 2010. Three lines of evidence to link outbreaks of the crown-of-thorns seastar Acanthaster planci to the release of larval food limitation. Coral Reefs 29 (3), 593–605. https://doi.org/10.1007/s00338-010-0628-z.

Farid, A.M., Mcfarlane, D.C., Farid, A.M., Mcfarlane, D.C., 2006. A Development of Degrees of Freedom for Manufacturing Systems.

Fayemi, P.E., Wanieck, K., Zollfrank, C., Maranzana, N., Aoussat, A., 2017. Biomimetics: process, tools and practice. Bioinspiration and Biomimetics 12 (1). https://doi.org/10.1088/1748-3190/12/1/011002. Institute of Physics Publishing, Feb. 01.

Ferrera, Maxime, Arnaubec, Aurelien, Istenic, Klemen, Gracias, Nuno, Bajjouk, Touria, 2021. Hyperspectral 3D Mapping of Underwater Environments, pp. 3696–3705. https://doi.org/10.1109/ICCVW54120.2021.00413.

Fine, M., Meroz-Fine, E., Hoegh-Guldberg, O., Jan. 2005. Tolerance of endolithic algae to elevated temperature and light in the coral Montipora monasteriata from the southern Great Barrier Reef. J. Exp. Biol. 208 (1), 75–81. https://doi.org/10.1242/jeb.01381.

Finney, S., Stephen, J., 2005. Photo Mosaics in Shallow Water Environments: Challenges and Results [Online]. Available: www.witpress.com.

Friedman, A., Pizarro, O., Williams, S.B., 2010. Rugosity, slope and aspect from bathymetric stereo image reconstructions. In: OCEANS'10 IEEE Sydney, OCEANSSYD 2010. https://doi.org/10.1109/OCEANSSYD.2010.5604003.

Gao, C., Li, P., Zhang, Y., Liu, J., Wang, L., Oct. 2016. People counting based on head detection combining Adaboost and CNN in crowded surveillance environment. Neurocomputing 208, 108–116. https://doi.org/10.1016/j.neucom.2016.01.097.

Gao, Z., Shi, Q., Fukuda, T., Li, C., Huang, Q., Mar. 2019. An overview of biomimetic robots with animal behaviors. Neurocomputing 332, 339–350. https://doi.org/10.1016/j.neucom.2018.12.071.

Gao, L., Li, P., Qin, H., Deng, Z., 2021. Mechatronic design and maneuverability analysis of a novel robotic shark for coral reef detection. In: Proceedings of the International Conference on Offshore Mechanics and Arctic Engineering - OMAE. https://doi.org/10.1115/OMAE2021-61080.

Garcia-Garcia, A., Orts-Escolano, S., Oprea, S., Villena-Martinez, V., Garcia-Rodriguez, J., Apr. 2017. A Review on Deep Learning Techniques Applied to Semantic Segmentation [Online]. Available: http://arxiv.org/abs/1704.06857.

Gardner, T.A., Côté, I.M., Gill, J.A., Grant, A., Watkinson, A.R., 2003. Long-Term Region-Wide Declines in Caribbean Corals.

Gastoldi, L., Cinti, S., Mar. 2023. (Bio)sensors applied to coral reefs' health monitoring: a critical overview. Green Anal. Chem. 4, 100049 https://doi.org/10.1016/J.GREEAC.2023.100049.

Giguere, P., Dudek, G., Prahacs, C., Plamondon, N., Turgeon, K., 2009. Unsupervised learning of terrain appearance for automated coral reef exploration. In: Proceedings of the 2009 Canadian Conference on Computer and Robot Vision, CRV 2009, pp. 268–275. https://doi.org/10.1109/CRV.2009.41.

Girdhar, Y., Dudek, G., Oct. 2016. Modeling curiosity in a mobile robot for long-term autonomous exploration and monitoring. Auton Robots 40 (7), 1267–1278. https://doi.org/10.1007/s10514-015-9500-x.

Girdhar, Y., et al., 2019. Streaming scene maps for co-robotic exploration in bandwidth limited environments. In: Proceedings-IEEE International Conference on Robotics and Automation, pp. 7940–7946. https://doi.org/10.1109/ICRA.2019.8794132.

Gleason, A.C.R., et al., 2010. Landscape video mosaic from a mesophotic coral reef. Coral Reefs 29 (2), 253. https://doi.org/10.1007/s00338-009-0544-2.

González-García, J., Gómez-Espinosa, A., Cuan-Urquizo, E., García-Valdovinos, L.G., Salgado-Jiménez, T., Escobedo Cabello, J.A., 2020. Autonomous underwater vehicles: Localization, navigation, and communication for collaborative missions. Appl. Sci. (Switzerland) 10 (4). https://doi.org/10.3390/app10041256. MDPI AG, Feb. 01.

González-Rivero, M., et al., Nov. 2014. The Catlin Seaview Survey - kilometre-scale seascape assessment, and monitoring of coral reef ecosystems. Aquat. Conserv. 24 (S2), 184–198. https://doi.org/10.1002/aqc.2505.

González-Rivero, M., et al., 2016. Scaling up ecological measurements of coral reefs using semi-automated field image collection and analysis. Remote Sens. (Basel) 8 (1). https://doi.org/10.3390/rs8010030.

González-Rivero, M., et al., Feb. 2020. Monitoring of coral reefs using artificial intelligence: a feasible and cost-effective approach. Remote Sens. (Basel) 12 (3), 489. https://doi.org/10.3390/rs12030489.

González-Sabbagh, S.P., Robles-Kelly, A., Jan. 2023. A survey on underwater computer vision. ACM Comput Surv. https://doi.org/10.1145/3578516.

Gouraguine, A., et al., Jan. 2019. Citizen science in data and resource-limited areas: a tool to detect long-term ecosystem changes. PloS One 14 (1). https://doi.org/10.1371/journal.pone.0210007.

Grabowski, R., Khosla, P., Choset, H., 2003. Autonomous exploration via regions of interest. In: IEEE International Conference on Intelligent Robots and Systems, pp. 1691–1696. https://doi.org/10.1109/iros.2003.1248887.

Grasmueck, M., Eberli, G.P., Viggiano, D.A., Correa, T., Rathwell, G., Luo, J., Dec. 2006. Autonomous underwater vehicle (AUV) mapping reveals coral mound distribution, morphology, and oceanography in deep water of the straits of Florida. Geophys. Res. Lett. 33 (23) https://doi.org/10.1029/2006GL027734.

Gray, A.E., et al., Dec. 2016. Comparison of reef fish survey data gathered by open and closed circuit SCUBA divers reveals differences in areas with higher fishing pressure. PloS One 11 (12). https://doi.org/10.1371/journal.pone.0167724.

Ha, D., Tang, Y., Aug. 2022. Collective intelligence for deep learning: a survey of recent developments. Collective Intell. 1 (1) https://doi.org/10.1177/26339137221114874, p. 263391372211148.

Hall, R., Skorina, E., Chiang, S.S., Onal, C.D., 2022. The effect of design and control parameters of a soft robotic fish tail to maximize propulsion force in undulatory actuation. In: Proceedings of the IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics. IEEE Computer Society. https://doi.org/10.1109/BioRob52689.2022.9925385.

Hammad, N.S., Khan, Z.N., Valle-Pérez, A.U., Hauser, C., 2023. A predictive machine learning model to optimize flow rates on an integrated microfluidic pumping system for peptide-based 3D bioprinting. SPIE-Intl Soc Optical Eng, Mar. 3. https://doi.org/10.1117/12.2650440.

Hedley, J.D., Roelfsema, C.M., Phinn, S.R., Mumby, P.J., Jan. 2012. Environmental and sensor limitations in optical remote sensing of coral reefs: implications for monitoring and sensor design. Remote Sens. (Basel) 4 (1), 271–302. https://doi.org/10.3390/rs4010271.

Hedley, J.D., et al., 2016a. Remote sensing of coral reefs for monitoring and management: a review. Remote Sensing 8 (2), 118, 2016, Vol. 8, Page 118. Feb. https://doi.org/10.3390/RS8020118.

Hedley, J.D., et al., 2016b. Remote sensing of coral reefs for monitoring and management: a review. Remote Sensing 8 (2). https://doi.org/10.3390/rs8020118. MDPI AG.

Henriques, J.F., Caseiro, R., Martins, P., Batista, J., Mar. 2015. High-speed tracking with kernelized correlation filters. IEEE Trans. Pattern Anal. Mach. Intell. 37 (3), 583–596. https://doi.org/10.1109/TPAMI.2014.2345390.

Hodgson, G., Liebler, J., 2002. The Global Coral Reef Crisis: Trends and Solutions.

Hoegh-Guldberg, O., 1999. Climate change, coral bleaching and the future of the world's coral reefs. Mar. Freshw. Res. 50 (8), 839–866. https://doi.org/10.1071/MF99078. CSIRO.

Hoeke, R., et al., Aug. 2009. Coral reef ecosystem integrated observing system: in-situ oceanographic observations at the US Pacific islands and atolls. J. Oper. Oceanogr. 2 (2), 3–14. https://doi.org/10.1080/1755876x.2009.12027737.

Hogue, A., Jenkin, M., 2006. Development of an underwater vision sensor for 3D reef mapping. In: IEEE International Conference on Intelligent Robots and Systems, pp. 5351–5356. https://doi.org/10.1109/IROS.2006.282097.

Hoogenboom, M.O., Campbell, D.A., Beraud, E., DeZeeuw, K., Ferrier-Pagès, C., Jan. 2012. Effects of light, food availability and temperature stress on the function of photosystem II and photosystem I of coral symbionts. PloS One 7 (1), e30167. https://doi.org/10.1371/journal.pone.0030167.

Hughes, T.P., et al., 2002. Detecting regional variation using meta-analysis and large-scale sampling: latitudinal patterns in recruitment. Ecology 83 (2), 436–451. https://doi.org/10.1890/0012-9658(2002)083[0436,DRVUMA]2.0.CO;2.

Hughes, T.P., et al., 2003. Climate Change, Human Impacts, and the Resilience of Coral Reefs.

Hughes, T.P., Graham, N.A.J., Jackson, J.B.C., Mumby, P.J., Steneck, R.S., Nov. 2010. Rising to the challenge of sustaining coral reef resilience. Trends Ecol. Evol. 25 (11), 633–642. https://doi.org/10.1016/J.TREE.2010.07.011.

IEEE Robotics and Automation Society and Institute of Electrical and Electronics Engineers, 2018. IEEE ROBIO 2018: 2018 IEEE International Conference on Robotics and Biomimetics: December 12–15. Kuala Lumpur, Malaysia.

IEEE Staff, 2016. OCEANS 2016 MTS IEEE Monterey. IEEE.

Jaffe, J. S., Moore, K. D., McLean Arete Associates, J, 2001. Underwater Optical Imaging: Status and Prospects, USA USA Michael Strand.

Jaffe, J.S., et al., Jan. 2017. A swarm of autonomous miniature underwater robot drifters for exploring submesoscale ocean dynamics. Nat. Commun. 8 https://doi.org/10.1038/ncomms14189.

Janzen, C.D., et al., 2019. Innovative real-time observing capabilities for remote coastal regions. Front. Mar. Sci. 6 (MAR) https://doi.org/10.3389/fmars.2019.00176. Frontiers Media S.A.

Joordens, M., Champion, B., 2015. Underwater swarm robotics: challenges and opportunities. In: Artificial Intelligence. IGI Global, pp. 2308–2332. https://doi.org/10.4018/978-1-5225-1759-7.ch096.

Kalantari, S., Nazemi, E., Masoumi, B., Jul. 2020. Emergence phenomena in self-organizing systems: a systematic literature review of concepts, researches, and future prospects. J. Organ. Comput. Electron. Commer. 30 (3), 224–265. https://doi.org/10.1080/10919392.2020.1748977.

Kalwa, J., et al., 2016. The european project MORPH: distributed UUV systems for multimodal, 3D underwater surveys. Mar. Technol. Soc. J. 50 (4), 26–41. https://doi.org/10.4031/MTSJ.50.4.10.

Katzschmann, R.K., Delpreto, J., Maccurdy, R., Rus, D., 2018a. Exploration of Underwater Life With an Acoustically Controlled Soft Robotic Fish [Online]. Available: https://www.science.org.

Katzschmann, R.K., DelPreto, J., MacCurdy, R., Rus, D., 2018b. Exploration of underwater life with an acoustically controlled soft robotic fish. Sci. Robot. 3 (16) https://doi.org/10.1126/SCIROBOTICS.AAR3449.

Kirchner, F., Straube, S., Kühn, D., Hoyer, N. (Eds.), 2020. AI Technology for Underwater Robots https://doi.org/10.1007/978-3-030-30683-0 [Online]. Available: http://www.springer.com/series/6259.

Knight, A., Cowling, R.M., Rouget, M., Balmford, A., 2008. Knowing but not doing: selecting priority conservation areas and the research-implementation gap. Conserv. Biol. https://doi.org/10.1111/j.1523-1739.2008.00914.x.

Knowlton, N., Jackson, J., 2008. Shifting Baselines, Local Impacts, and Global Change on Coral Reefs. PLoS Biology 6, e54. https://doi.org/10.1371/journal.pbio.0060054.

Koh, L.P., Wich, S.A., 2012. Dawn of drone ecology: low-cost autonomous aerial vehicles for conservation [Online]. Available: http://creativecommons.org/licenses/by/3.0/-The132.Availableonline:www.tropicalconservationscience.org.

Kushleyev, A., Mellinger, D., Powers, C., Kumar, V., Nov. 2013. Towards a swarm of agile micro quadrotors. Auton Robots 287–300. https://doi.org/10.1007/s10514-013-9349-9.

Lam, K., et al., May 2006. A comparison of video and point intercept transect methods for monitoring subtropical coral communities. J. Exp. Mar. Biol. Ecol. 333 (1), 115–128. https://doi.org/10.1016/j.jembe.2005.12.009.

Lam, K., Shin, P.K.S., Bradbeer, R., Ku, K.K.K., Hodgson, P., Cheung, S.G., 2007. Baseline data of subtropical coral communities in Hoi Ha Wan Marine Park, Hong Kong, obtained by an underwater remote operated vehicle (ROV). Mar. Pollut. Bull. 54 (1), 107–112. https://doi.org/10.1016/j.marpolbul.2006.09.023.

Lau, C.M., et al., Sep. 2019. Tracing coral reefs: a citizen science approach in mapping coral reefs to enhance marine park management strategies. Front. Mar. Sci. 6 https://doi.org/10.3389/fmars.2019.00539.

Lawrance, N., et al., May 2019. Shared autonomy for low-cost underwater vehicles. J. Field Robot 36 (3), 495–516. https://doi.org/10.1002/rob.21835.

Leon, J.X., Roelfsema, C.M., Saunders, M.I., Phinn, S.R., Aug. 2015. Measuring coral reef terrain roughness using 'structure-from-motion' close-range photogrammetry. Geomorphology 242, 21–28. https://doi.org/10.1016/j.geomorph.2015.01.030.

Leonard, N.E., Paley, D.A., Davis, R.E., Fratantoni, D.M., Lekien, F., Zhang, F., Nov. 2010. Coordinated control of an underwater glider fleet in an adaptive ocean sampling field experiment in Monterey bay. J. Field Robot. 27 (6), 718–740. https://doi.org/10.1002/rob.20366.

Li, P., Wang, D., Wang, L., Lu, H., Apr. 2018. Deep visual tracking: review and experimental comparison. Pattern Recogn. 76, 323–338. https://doi.org/10.1016/j.patcog.2017.11.007.

Lin, T.Y., Goyal, P., Girshick, R., He, K., Dollar, P., Feb. 2020. Focal loss for dense object detection. IEEE Trans. Pattern Anal. Mach. Intell. 42 (2), 318–327. https://doi.org/10.1109/TPAMI.2018.2858826.

Loya, Y., Puglise, K.A., Bridge, T.C.L., 2019. Mesophotic Coral Ecosystems [Online]. Available: http://www.springer.com/series/7539.

Luong, J., et al., 2019. Eversion and Retraction of a Soft Robot Towards the Exploration of Coral Reefs; Eversion and Retraction of a Soft Robot Towards the Exploration of Coral Reefs (doi: 10.0/Linux-x86_64).

Macleod, Miles, Nagatsu, Michiru, 2018. What does interdisciplinarity look like in practice: mapping interdisciplinarity and its limits in the environmental sciences. Stud. Hist. Philos. Sci. A 67. https://doi.org/10.1016/j.shpsa.2018.01.001.

Madin, E.M.P., Darling, E.S., Hardt, M.J., Dec. 2019. Emerging technologies and coral reef conservation: opportunities, challenges, and moving forward. Front. Mar. Sci. 6 https://doi.org/10.3389/fmars.2019.00727.

Mahmood, A., et al., Aug. 2016. Coral classification with hybrid feature representations. In: Proceedings - International Conference on Image Processing, ICIP. IEEE Computer Society, pp. 519–523. https://doi.org/10.1109/ICIP.2016.7532411.

Manderson, T., Li, J., Poza, D.C., Dudek, N., Meger, D., Dudek, G., 2016. Towards Autonomous Robotic Coral Reef Health Assessment, 113. https://doi.org/10.1007/978-3-319-27702-8_7.

Manderson, T., Li, J., Dudek, N., Meger, D., Dudek, G., 2017a. Robotic coral reef health assessment using automated image analysis. J. Field Robot 34 (1), 170–187. https://doi.org/10.1002/rob.21698.

Manderson, T., Li, J., Dudek, N., Meger, D., Dudek, G., 2017b. Robotic coral reef health assessment using automated image analysis. J. Field Robot 34 (1), 170–187. Jan. https://doi.org/10.1002/rob.21698.

Mary, N. Ani Brown, Dharma, D., 2017. Coral reef image classification employing Improved LDP for feature extraction. J. Visual Commun. Image Represent. 49, 225–242. Academic Press Inc.. Nov. 01. https://doi.org/10.1016/j.jvcir.2017.09.008.

Maslin, M., Louis, S., Godary Dejean, K., Lapierre, L., Villéger, S., Claverie, T., Dec. 2021. Underwater robots provide similar fish biodiversity assessments as divers on coral reefs. Remote Sens Ecol Conserv 7 (4), 567–578. https://doi.org/10.1002/rse2.209.

McCook, L.J., et al., 2010. Adaptive management of the Great Barrier Reef: a globally significant demonstration of the benefits of networks of marine reserves. Proc. Natl. Acad. Sci. U. S. A. 107 (43), 18278–18285. Oct. 26. https://doi.org/10.1073/pnas.0909335107.

Mellin, C., Peterson, E.E., Puotinen, M., Schaffelke, B., Sep. 2020. Representation and complementarity of the long-term coral monitoring on the great barrier reef. Ecol. Appl. 30 (6) https://doi.org/10.1002/eap.2122.

Mills, M., Pressey, R.L., Weeks, R., Foale, S., Ban, N.C., Sep. 2010. A mismatch of scales: challenges in planning for implementation of marine protected areas in the Coral Triangle. Conserv. Lett. 3 (5), 291–303. https://doi.org/10.1111/j.1755-263X.2010.00134.x.

Modasshir, M., Rekleitis, I., 2020. Enhancing coral reef monitoring utilizing a deep semi-supervised learning approach. In: Proceedings - IEEE International Conference on Robotics and Automation, pp. 1874–1880. https://doi.org/10.1109/ICRA40945.2020.9196528.

Modasshir, M., Quattrini Li, A., Rekleitis, I., 2018a. Deep neural networks: A comparison on different computing platforms. In: Proceedings - 2018 15th Conference on Computer and Robot Vision, CRV 2018. Institute of Electrical and Electronics Engineers Inc., pp. 383–389. https://doi.org/10.1109/CRV.2018.00060. Dec.

Modasshir, M., Rahman, S., Youngquist, O., Rekleitis, I., 2018b. Coral identification and counting with an autonomous underwater vehicle. In: 2018 IEEE International Conference on Robotics and Biomimetics (ROBIO). IEEE, pp. 524–529.

Modasshir, M., Rahman, S., Rekleitis, I., 2021. Autonomous 3d semantic mapping of coral reefs. In: Field and Service Robotics: Results of the 12th International Conference. Springer, pp. 365–379.

Moniruzzaman, M., Islam, S.M.S., Bennamoun, M., Lavery, P., 2017. Deep learning on underwater marine object detection: A survey. In: Advanced Concepts for Intelligent Vision Systems: 18th International Conference, ACIVS 2017. Springer, pp. 150–160. Antwerp, Belgium, September 18–21, 2017, Proceedings 18.
Morais, J., Medeiros, A.P.M., Santos, B.A., Sep. 2018. Research gaps of coral ecology in a changing world. Mar. Environ. Res. 140, 243–250. https://doi.org/10.1016/J.MARENVRES.2018.06.021.
Mountrakis, G., Im, J., Ogole, C., May 2011. Support vector machines in remote sensing: a review. ISPRS J. Photogramm. Remote Sens. 66 (3), 247–259. https://doi.org/10.1016/j.isprsjprs.2010.11.001.
Muhlenbach, F., Zighed, D.A., 2004. Identifying and Handling Mislabelled Instances.
Mulhall, M., 2004. Saving the Rainforests of the Sea: An Analysis of International Efforts to Conserve Coral Reefs [Online]. Available: http://www.fao.org/newsroom/en/news/2005/102911/index.html.
Muller-Karger, F.E., et al., 2018. Advancing marine biological observations and data requirements of the complementary Essential Ocean Variables (EOVs) and Essential Biodiversity Variables (EBVs) frameworks. Front. Mar. Sci. 5 (JUN) https://doi.org/10.3389/fmars.2018.00211. Frontiers Media S.A., Jun. 27.
Mumby, P.J., et al., 2004. Remote sensing of coral reefs and their physical environment. Mar. Pollut. Bull. 48 (3–4), 219–228. https://doi.org/10.1016/j.marpolbul.2003.10.031.
Mumby, P.J., Hastings, A., Edwards, H.J., Nov. 2007. Thresholds and the resilience of Caribbean coral reefs. Nature 450 (7166), 98–101. https://doi.org/10.1038/nature06252.
Munn, Z., Peters, M.D.J., Stern, C., Tufanaru, C., McArthur, A., Aromataris, E., Nov. 2018. Systematic review or scoping review? Guidance for authors when choosing between a systematic or scoping review approach. BMC Med. Res. Methodol. 18 (1) https://doi.org/10.1186/s12874-018-0611-x.
Murad, M.F.A.B.M., Samah, M.I.B., Ismail, Z.H., Sammut, K., 2016. Surveillance of coral reef development using an autonomous underwater vehicle. In: Autonomous Underwater Vehicles 2016, pp. 14–19. https://doi.org/10.1109/AUV.2016.7778713. AUV 2016.
Nguyen, H.T., Lee, E.H., Lee, S., Jan. 2020. Study on the classification performance of underwater sonar image classification based on convolutional neural networks for detecting a submerged human body. Sensors (Switzerland) 20 (1). https://doi.org/10.3390/s20010094.
Nicholson, J.W., Healey, A.J., 2008. The present state of autonomous underwater vehicle (AUV) applications and technologies. Mar. Technol. Soc. J. 42, 44–51. https://doi.org/10.4031/002533208786861272.
Ninio, R., Delean, S., Osborne, K., Sweatman, H., 2003. Estimating cover of benthic organisms from underwater video images variability associated with multiple observers. https://doi.org/10.3354/meps265107.
Obura, D.O., et al., 2019. Coral reef monitoring, reef assessment technologies, and ecosystem-based management. Front. Mar. Sci. 6 (SEP) https://doi.org/10.3389/fmars.2019.00580.
Osborne, K., Dolman, A.M., Burgess, S.C., Johns, K.A., 2011. Disturbance and the dynamics of coral cover on the Great Barrier Reef (1995–2009). PloS One 6 (3). https://doi.org/10.1371/journal.pone.0017516.
Pante, E., Dustan, P., 2012. Getting to the point: accuracy of point count in monitoring ecosystem change. J. Mar. Biol. 2012, 1–7. https://doi.org/10.1155/2012/802875.
Parmesan, C., Yohe, G., 2003. A Globally Coherent Fingerprint of Climate Change Impacts Across Natural Systems.
Pham, V.-Q., Kozakaya, T., Yamaguchi, O., Okada, R., Dec. 2015. COUNT forest: CO-voting uncertain number of targets using random forest for crowd density estimation. In: 2015 IEEE International Conference on Computer Vision (ICCV). IEEE, pp. 3253–3261. https://doi.org/10.1109/ICCV.2015.372.
Pizarro, O., Friedman, A., Bryson, M., Williams, S.B., Madin, J., Mar. 2017. A simple, fast, and repeatable survey method for underwater visual 3D benthic mapping and monitoring. Ecol. Evol. 7 (6), 1770–1782. https://doi.org/10.1002/ece3.2701.
Poongundran, M., Prasannan, D., Agnes, J.S., Das, S., Anusha, D.J., Amandykova, D., 2022. Role of underwater robots in ocean exploration research. In: Proceedings - International Conference on Applied Artificial Intelligence and Computing, ICAAIC 2022. Institute of Electrical and Electronics Engineers Inc., pp. 1789–1794. https://doi.org/10.1109/ICAAIC53929.2022.9793261
Preiss, J.A., Honig, W., Sukhatme, G.S., Ayanian, N., Jul. 2017. Crazyswarm: a large nano-quadcopter swarm. In: Proceedings - IEEE International Conference on Robotics and Automation. Institute of Electrical and Electronics Engineers Inc., pp. 3299–3304. https://doi.org/10.1109/ICRA.2017.7989376
Prestero, T., 2001. Development of a six-degree of freedom simulation model for the REMUS autonomous underwater vehicle. Oceans Conf. Rec. (IEEE) 1, 450–455. https://doi.org/10.1109/OCEANS.2001.968766.
Redmon, J., Divvala, S., Girshick, R., Farhadi, A., Dec. 2016. You only look once: Unified, real-time object detection. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition. IEEE Computer Society, pp. 779–788. https://doi.org/10.1109/CVPR.2016.91.
Roelfsema, C., Apr. 2010. Integrating field data with high spatial resolution multispectral satellite imagery for calibration and validation of coral reef benthic community maps. J. Appl. Remote. Sens. 4 (1), 043527 https://doi.org/10.1117/1.3430107.
Rogers, C.S., Garrison, G., Grober, R., Hillis, Z.-M., Franke, M.A., 1994. Coral Reef Monitoring Manual for the Caribbean and Western Atlantic Coral Reef Monitoring Manual for the Caribbean and Western Atlantic CONTENTS FOREWORD Purpose Acknowledgements Photo and Illustration Credits.
Ruiz, S., Pascual, A., Garau, B., Faugère, Y., Alvarez, A., Tintoré, J., 2009. Mesoscale dynamics of the Balearic Front, integrating glider, ship and satellite data. J. Mar. Syst. 78 (Suppl. 1) https://doi.org/10.1016/j.jmarsys.2009.01.007.
Safuan, C.D.M., et al., Sep. 2022. Assessment of coral health status using two-dimensional Coral Health Index (2D-CHI): a preliminary study in Pulau Perhentian Marine Park, Malaysia. Reg. Stud. Mar. Sci. 55, 102543 https://doi.org/10.1016/J.RSMA.2022.102543.
Sahoo, A., Dwivedy, S.K., Robi, P.S., 2019. Advancements in the field of autonomous underwater vehicle. In: Ocean Engineering, 181. Elsevier Ltd, pp. 145–160. Jun. 01. https://doi.org/10.1016/j.oceaneng.2019.04.011.
Sam, D.B., Surya, S., Babu, R.V., Nov. 2017. Switching convolutional neural network for crowd counting. In: Proceedings - 30th IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2017. Institute of Electrical and Electronics Engineers Inc., pp. 4031–4039. https://doi.org/10.1109/CVPR.2017.429
Sánchez-Ferreira, C., Mori, J.Y., Farias, M.C.Q., Llanos, C.H., Oct. 2016. A real-time stereo vision system for distance measurement and underwater image restoration. J. Braz. Soc. Mech. Sci. Eng. 38 (7), 2039–2049. https://doi.org/10.1007/s40430-016-0596-5.
Santos-Victor, J., Sandini, G., 1997. Visual Behaviors for Docking*.
Sattar, J., Giguëre, P., Dudek, G., 2009. Sensor-based behavior control for an autonomous underwater vehicle. Int. J. Robot. Res. 28 (6), 701–713. https://doi.org/10.1177/0278364908098560.
Schmickl, T., et al., 2011. CoCoRo - The self-aware underwater swarm. In: Proceedings - 2011 5th IEEE Conference on Self-Adaptive and Self-Organizing Systems Workshops, SASOW 2011, pp. 120–126. https://doi.org/10.1109/SASOW.2011.11.
Schutte, V.G.W., Selig, E.R., Bruno, J.F., Mar. 2010. Regional spatio-temporal trends in Caribbean coral reef benthic communities. Mar. Ecol. Prog. Ser. 402, 115–122. https://doi.org/10.3354/meps08438.
Scopélitis, J., Andréfouët, S., Largouët, C., Jul. 2007. Modelling coral reef habitat trajectories: evaluation of an integrated timed automata and remote sensing approach. Ecol. Model. 205 (1–2), 59–80. https://doi.org/10.1016/j.ecolmodel.2007.02.011.
Shackleton, C.M., Cundill, G., Knight, A.T., 2009. Beyond just research: experiences from southern Africa in developing social learning partnerships for resource conservation initiatives. Biotropica 41 (5).
Shihavuddin, A.S.M., Gracias, N., Garcia, R., Gleason, A.C.R., Gintert, B., Apr. 2013. Image-based coral reef classification and thematic mapping. Remote Sens. (Basel) 5 (4), 1809–1841. https://doi.org/10.3390/rs5041809.
Shkurti, F., et al., 2012. Multi-domain monitoring of marine environments using a heterogeneous robot team. In: IEEE International Conference on Intelligent Robots and Systems, pp. 1747–1753. https://doi.org/10.1109/IROS.2012.6385685.
Shrivastava, A., Gupta, A., Girshick, R., Dec. 2016. Training region-based object detectors with online hard example mining. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition. IEEE Computer Society, pp. 761–769. https://doi.org/10.1109/CVPR.2016.89.
Snow, C., Zeng, J., Nonas-Hunter, L., Diggins, D., Bennett, K., Bennett, A., Oct. 2020. Design of a cable-mounted robot for near shore monitoring. In: 2020 Global Oceans 2020: Singapore - U.S. Gulf Coast. Institute of Electrical and Electronics Engineers Inc. https://doi.org/10.1109/IEEECONF38699.2020.9389423
Szostak, R., 2013. The state of the field: interdisciplinary research. In: Issues in Interdisciplinary Studies [Preprint]. https://doi.org/10.7939/R3QB9V49Q.
Teague, J., Megson-Smith, D.A., Allen, M.J., Day, J.C.C., Scott, T.B., Jan. 2022. A review of current and new optical techniques for coral monitoring. Oceans 3 (1), 30–45. https://doi.org/10.3390/oceans3010003.
Rädecker, N., Pogoreutz, C., Gegner, H., Cardenas, A., Roth, F., Bougoure, J., Guagliardo, P., Wild, C., Pernice, M., Raina, J., Meibom, A., Voolstra, C., 2021. Heat stress destabilizes symbiotic nutrient cycling in corals. Proceedings of the National Academy of Sciences. 118. e2022653118. https://doi.org/10.1073/pnas.2022653118/-/DCSupplemental.
Singh, H., et al., 2004. Imaging Coral I: Imaging Coral Habitats with the SeaBED AUV.
UK-RAS Network Robotics & Autonomous Systems, 2018. Agricultural Robotics: The Future of Robotic Agriculture [Online]. Available: www.ukras.org.
Van Der Zwaan, S., Bernardino, A., Santos-Victor, J., 2002. Visual Station Keeping for Floating Robots in Unstructured Environments [Online]. Available: www.isr.ist.
Van Hooidonk, R., Maynard, J.A., Planes, S., May 2013. Temporary refugia for coral reefs in a warming world. Nat. Clim. Chang. 3 (5), 508–511. https://doi.org/10.1038/nclimate1829.
Villon, S., et al., Nov. 2018. A Deep learning method for accurate and fast identification of coral reef fishes in underwater images. Ecol. Inform. 48, 238–244. https://doi.org/10.1016/j.ecoinf.2018.09.007.
Visbeck, M., 2018. Ocean science research is key for a sustainable future. Nat. Commun. 9 (1) https://doi.org/10.1038/s41467-018-03158-3. Nature Publishing Group, Dec. 01.
Voolstra, C.R., 2020. Shining light on deep-sea bioluminescence. Editorial comment on the highlight article 'Biochemical characterization of diverse deep-sea anthozoan bioluminescence systems' by M. Besso-Uehara et al. 2020. Marine Biology 167 (8). https://doi.org/10.1007/s00227-020-03732-8. Springer, Aug. 01.
Wang, R., Wang, S., Wang, Y., Cheng, L., Tan, M., Feb. 2022. Development and motion control of biomimetic underwater robots: a survey. IEEE Trans Syst Man Cybern Syst 52 (2), 833–844. https://doi.org/10.1109/TSMC.2020.3004862.
Wangpraseurt, D., You, S., Sun, Y., Chen, S., T. authors contribute equally, 2022. Biomimetic 3D Living Materials Powered by Microorganisms [Online]. Available: https://orcid.org/0000-0003-4834-8981.
Weinstein, A., Cho, A., Loianno, G., Kumar, V., Jul. 2018. Visual inertial odometry swarm: an autonomous swarm of vision-based quadrotors. IEEE Robot. Autom. Lett. 3 (3), 1801–1807. https://doi.org/10.1109/LRA.2018.2800119.
Wetz, J.J., Ajemian, M.J., Shipley, B., Stunz, G.W., May 2020. An assessment of two visual survey methods for documenting fish community structure on artificial

platform reefs in the Gulf of Mexico. Fish. Res. 225, 105492 https://doi.org/10.1016/J.FISHRES.2020.105492.

Whitcomb, L., Yoerger, D.R., Singh, H., Howland, J., 2000. Advances in underwater robot vehicles for deep ocean exploration: navigation, control, and survey operations. In: Robotics Research: The Ninth International Symposium. Springer, pp. 439–448.

Venkatesh, C., Sekhar, C.C., Venugopal, M., Kumar, M.B.P., 2013. A robotic colonizer system for identifying and estimation of marine habitats in aqueous environment. In: 2013 International Conference on Optical Imaging Sensor and Security (ICOSS), Coimbatore, India, 2013, pp. 1–5. https://doi.org/10.1109/ICOISS.2013.6678414.

Wilson, A.D.M., Wikelski, M., Wilson, R.P., Cooke, S.J., 2015. Utility of biological sensor tags in animal conservation. Conserv. Biol. 29 (4), 1065–1075. Aug. 01. https://doi.org/10.1111/cobi.12486.

Wölfl, A.C., et al., 2019. Seafloor mapping - the challenge of a truly global ocean bathymetry. Front. Mar. Sci. 6 (JUN) https://doi.org/10.3389/fmars.2019.00283. Frontiers Media S.A.

Woodhead, A.J., Hicks, C.C., Norström, A.V., Williams, G.J., Graham, N.A.J., Jun. 2019. Coral reef ecosystem services in the Anthropocene. Funct. Ecol. 33 (6), 1023–1034. https://doi.org/10.1111/1365-2435.13331.

Wooldridge, S., Done, T., Apr. 2004. Learning to predict large-scale coral bleaching from past events: a Bayesian approach using remotely sensed data, in-situ data, and environmental proxies. Coral Reefs 23 (1), 96–108. https://doi.org/10.1007/s00338-003-0361-y.

Xu, B., Qiu, G., May 2016. Crowd density estimation based on rich features and random projection forest. In: 2016 IEEE Winter Conference on Applications of Computer Vision, WACV 2016. Institute of Electrical and Electronics Engineers Inc. https://doi.org/10.1109/WACV.2016.7477682

Yuh, J., 2000. Design and Control of Autonomous Underwater Robots: A Survey [Online]. Available: http://www.eng.hawaii.edu/.

Zahadat, P., Schmickl, T., Apr. 2016. Division of labor in a swarm of autonomous underwater robots by improved partitioning social inhibition. Adapt. Behav. 24 (2), 87–101. https://doi.org/10.1177/1059712316633028.

Zamani, N.P., Zuhdi, M.F., Madduppa, H., Aug. 2022. Environmental DNA biomonitoring reveals seasonal patterns in coral reef fish community structure. Environ. Biol. Fishes 105 (8), 971–991. https://doi.org/10.1007/s10641-022-01274-0.

Zantis, L.J., Carroll, E.L., Nelms, S.E., Bosker, T., 2021. Marine mammals and microplastics: a systematic review and call for standardisation. Environ. Pollut. 269 https://doi.org/10.1016/j.envpol.2020.116142. Elsevier Ltd, Jan. 15.

Zhang, H., Negahdaripour, S., 2003. On reconstruction of 3-D volumetric models of reefs and benthic structures from image sequences of a stereo rig. In: Oceans Conference Record (IEEE), pp. 2553–2559.

Zhang, Y., Zhou, D., Chen, S., Gao, S., Ma, Y., Dec. 2016. Single-image crowd counting via multi-column convolutional neural network. In: Proceedings of the IEEE Computer Society Conference on Computer Vision and Pattern Recognition. IEEE Computer Society, pp. 589–597. https://doi.org/10.1109/CVPR.2016.70.

Ziegler, M., et al., Mar. 2017. Biogeography and molecular diversity of coral symbionts in the genus Symbiodinium around the Arabian Peninsula. J. Biogeogr. 44 (3), 674–686. https://doi.org/10.1111/jbi.12913.

Zweifler, A., Akkaynak, D., Mass, T., Treibitz, T., Sep. 2017. In situ analysis of coral recruits using fluorescence imaging. Front. Mar. Sci. 4 (SEP) https://doi.org/10.3389/fmars.2017.00273.